

Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 14

Imaging of the Abdomen and Pelvis Fall 2025

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Recommended Reading

- Drake, Vogl, Mitchell. Gray's Anatomy for Students. 4th ed. Elsevier Churchill Livingstone. Chapters 4 and 5.
- *Pay special attention to the “In the Clinic” boxes for the clinical correlations for this topic*

Objectives

List the parietal and visceral branches of the abdominal aorta and use this information to identify them on cadaveric specimen and imaging studies.	SOM.MK.2.BPM1.1.MSK.1.ANAT.1268
List the approximate vertebral levels of the visceral branches of the abdominal aorta and its bifurcation into the common iliac arteries and their distribution.	SOM.MK.2.BPM1.1.MSK.1.ANAT.1269
Identify the bones, muscles and vessels of the anterior/posterior abdominal wall and pelvic wall on imaging studies.	SOM.MK.2.BPM1.1.MSK.1.ANAT.1289
Describe types of pelvic fractures based on mechanism of injury.	SOM.MKII.BPM1.1.MSK.1.ANAT.1495

Overview

- Abdominal Aorta
 - Branches
 - Imaging
- Imaging of the anterior/posterior abdominal wall and pelvic wall
 - Axial CTs
 - Coronal CTs
- Pelvic Fractures

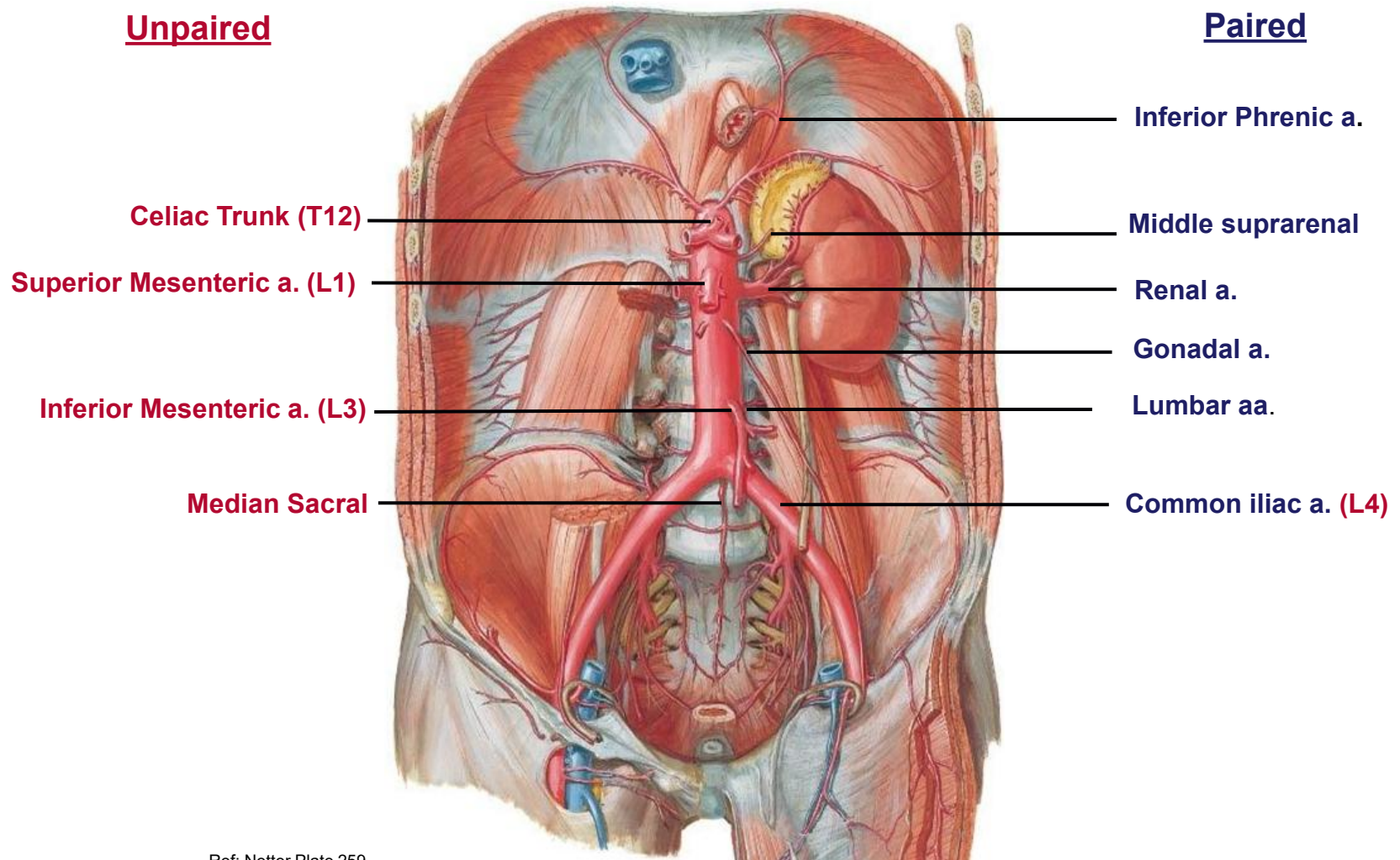
Abdominal Aorta

- Continuation of the thoracic aorta.
- **Passes through the diaphragm at the T12 vertebral level.**
- Gives paired and unpaired branches to both the abdominal viscera and abdominal wall.
- **Bifurcates at L4** into the common iliac arteries.
- Common iliac arteries divide into internal and external iliac arteries.
- The **internal iliac artery supplies the pelvis and gluteal region**
- The **external iliac artery becomes the femoral artery** as it passes deep to the inguinal ligament and supplies the lower limb.

Abdominal Aorta - Parietal and Visceral branches

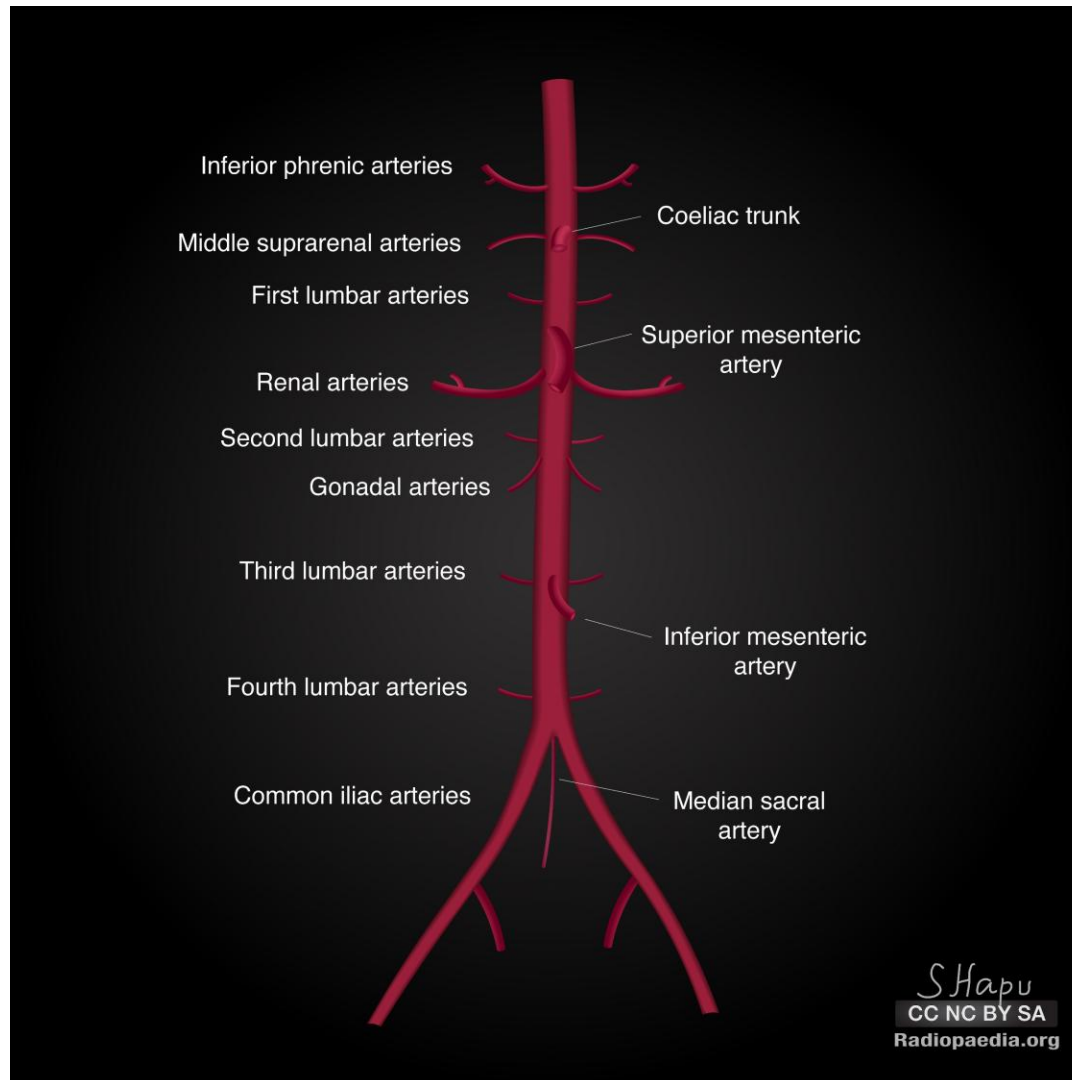
Unpaired

Paired

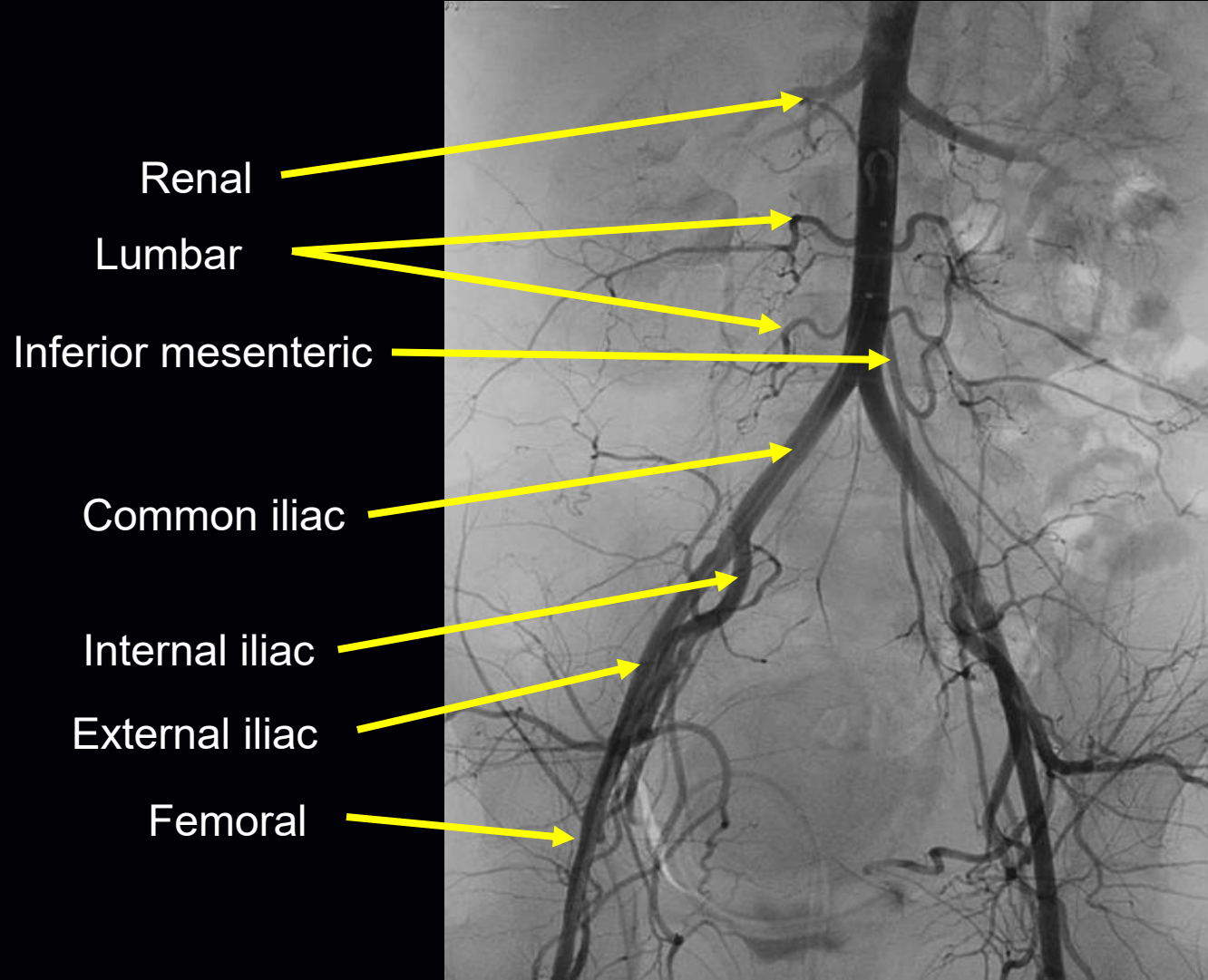


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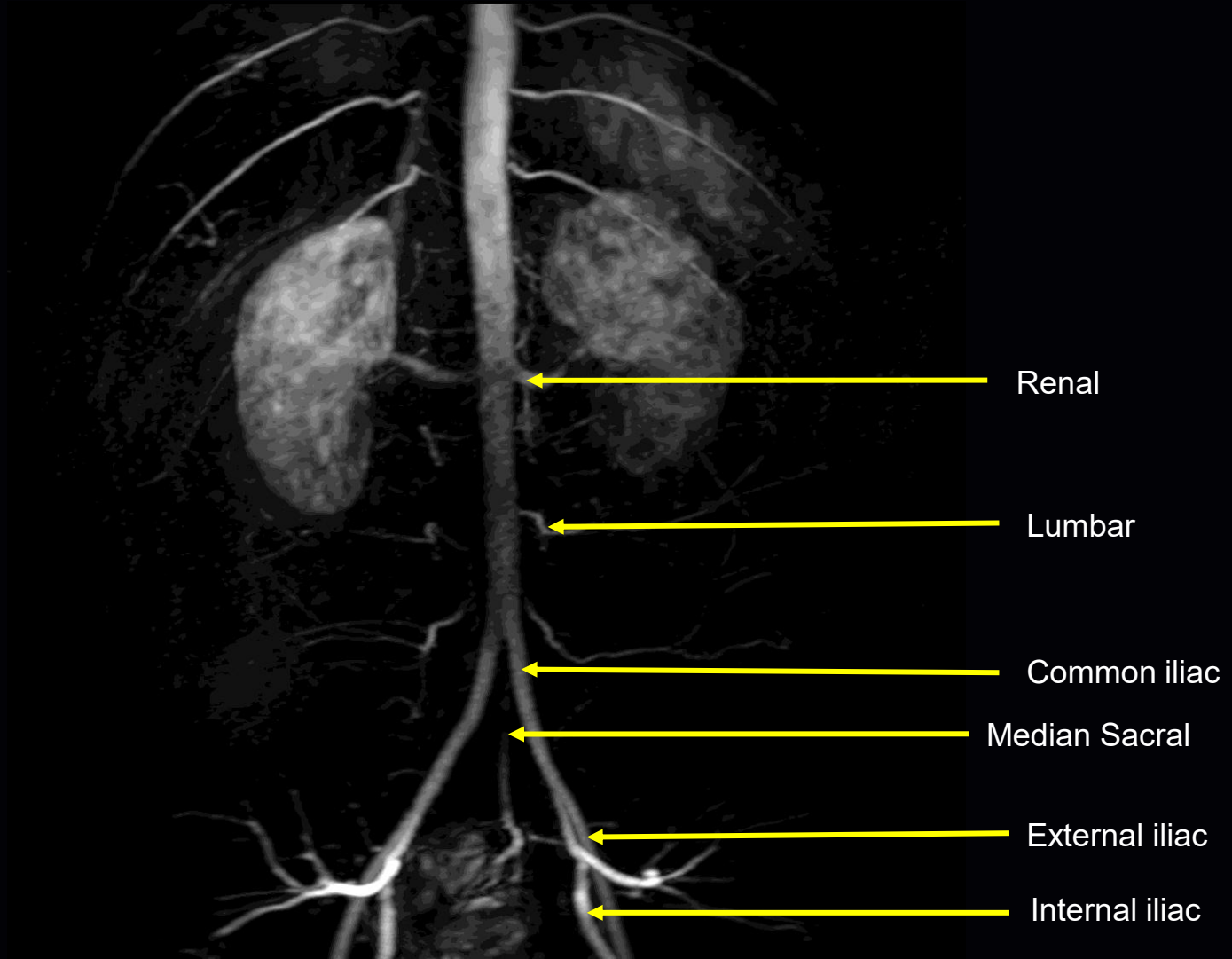
Abdominal aorta (illustration)



Abdominal Aorta – AP Arteriogram (Normal)

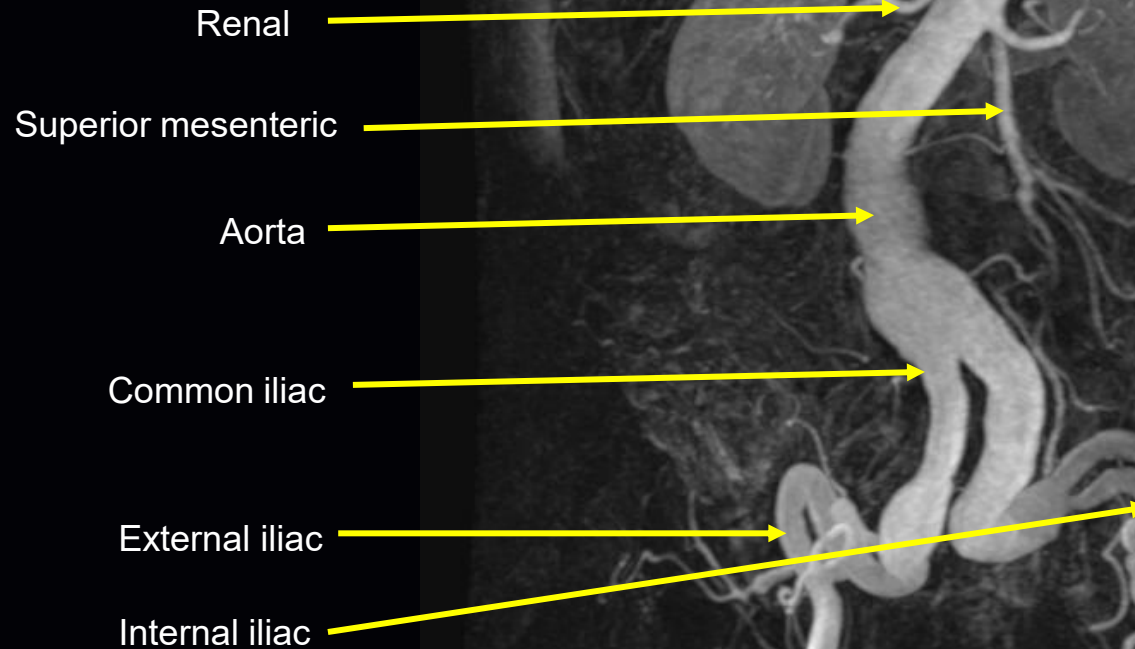


Abdominal Aorta - Coronal MRA (Normal)



Abdominal Aorta - Coronal MRA (Abnormal)

70-year-old female with pulsating mass in her right abdomen



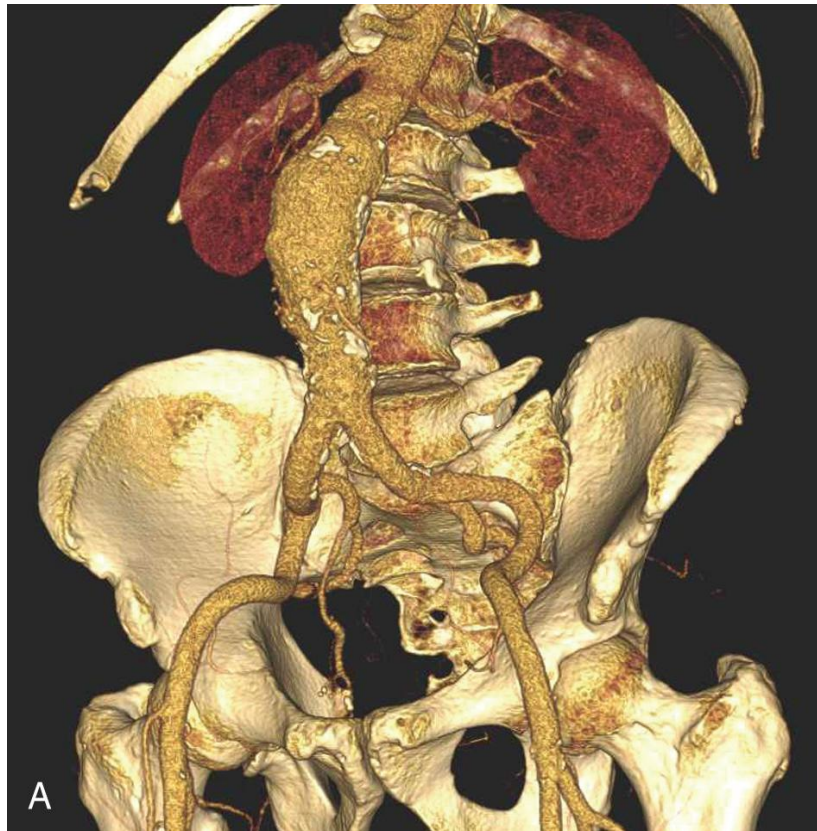
<https://radiopaedia.org/cases/17246>

Dilation of the abdominal aorta and iliac arteries

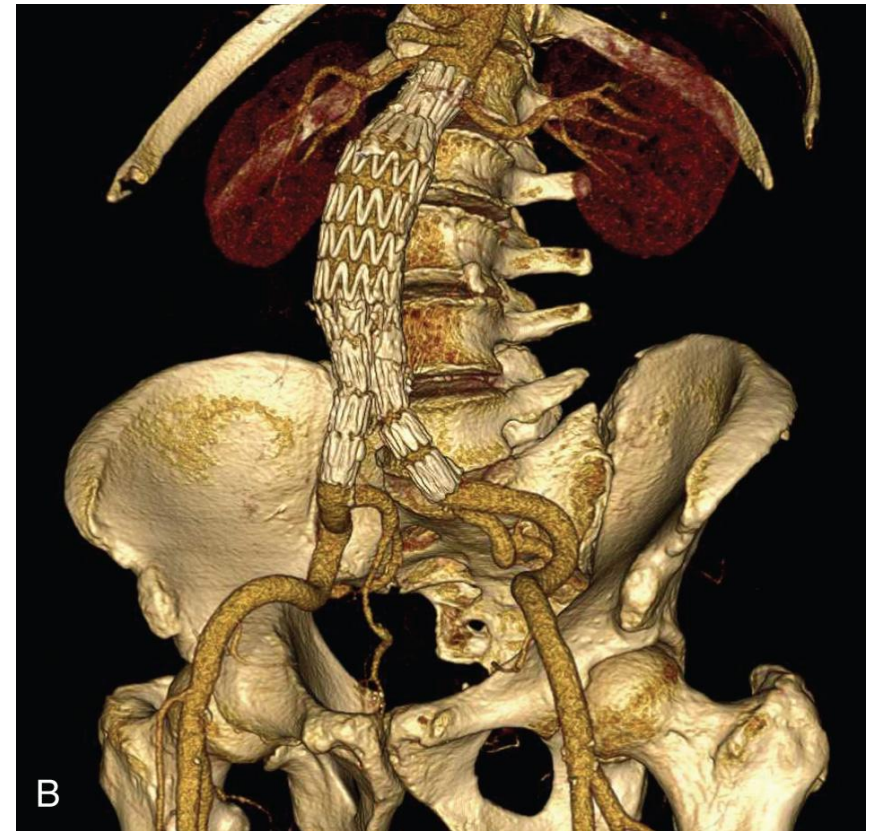
Abdominal Aortic Aneurysm (AAA)

- Dilation of the abdominal aorta (Most commonly infrarenal).
- Cases are rising due to an **aging** population and **increased imaging** usage (detecting asymptomatic patients).
- As the aorta expands, rupture risk increases.
- Operations are typically considered for aneurysms **5.5 cm** or larger.
- Traditional treatment involved an **open operative technique** involving a large incision from the xiphoid process to the symphysis pubis.
- New treatments include the **endovascular graft.**

Volume-rendered reconstruction using multidetector CT scan



Abdominal aortic aneurysm

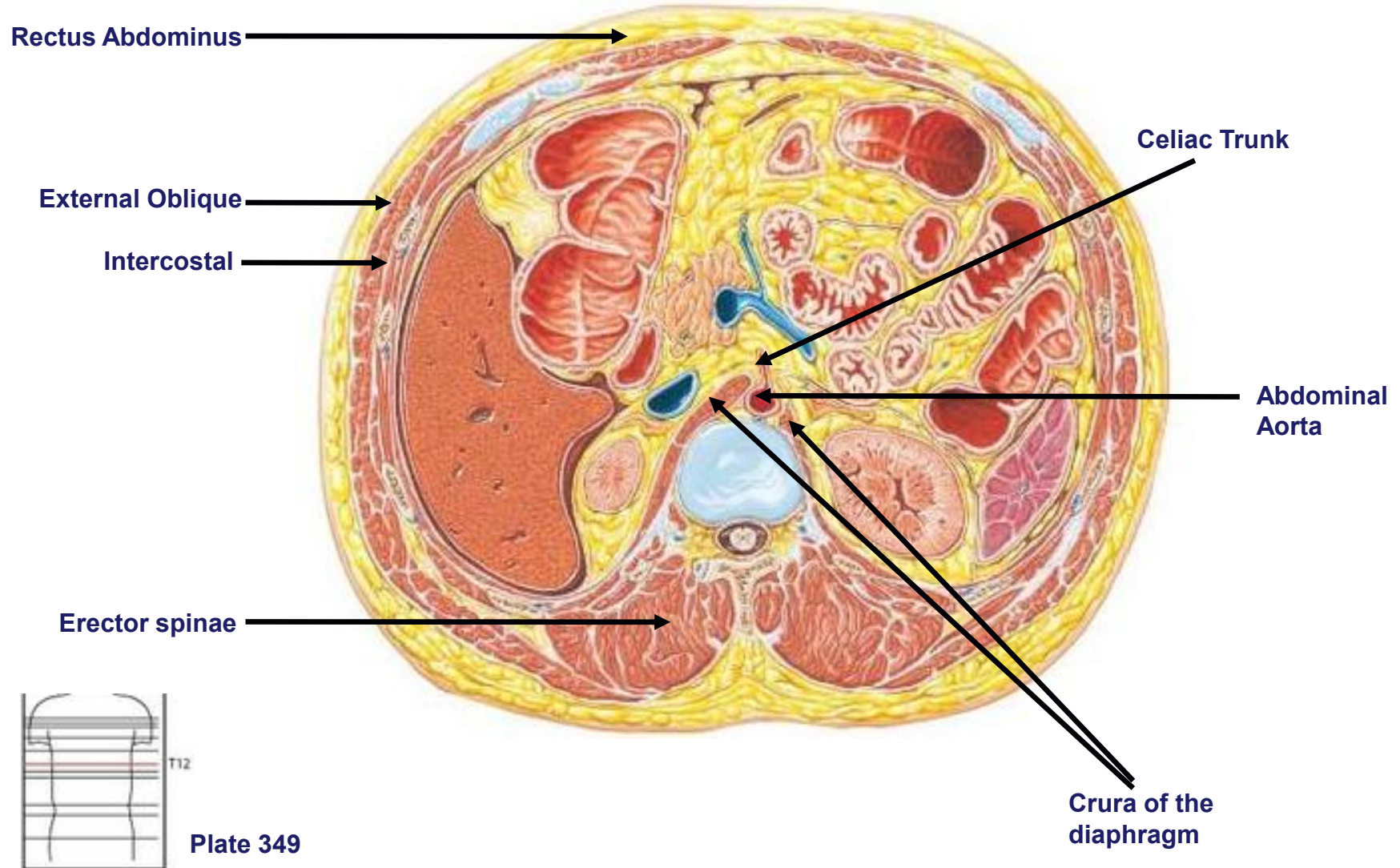


Endovascular aneurysm repair

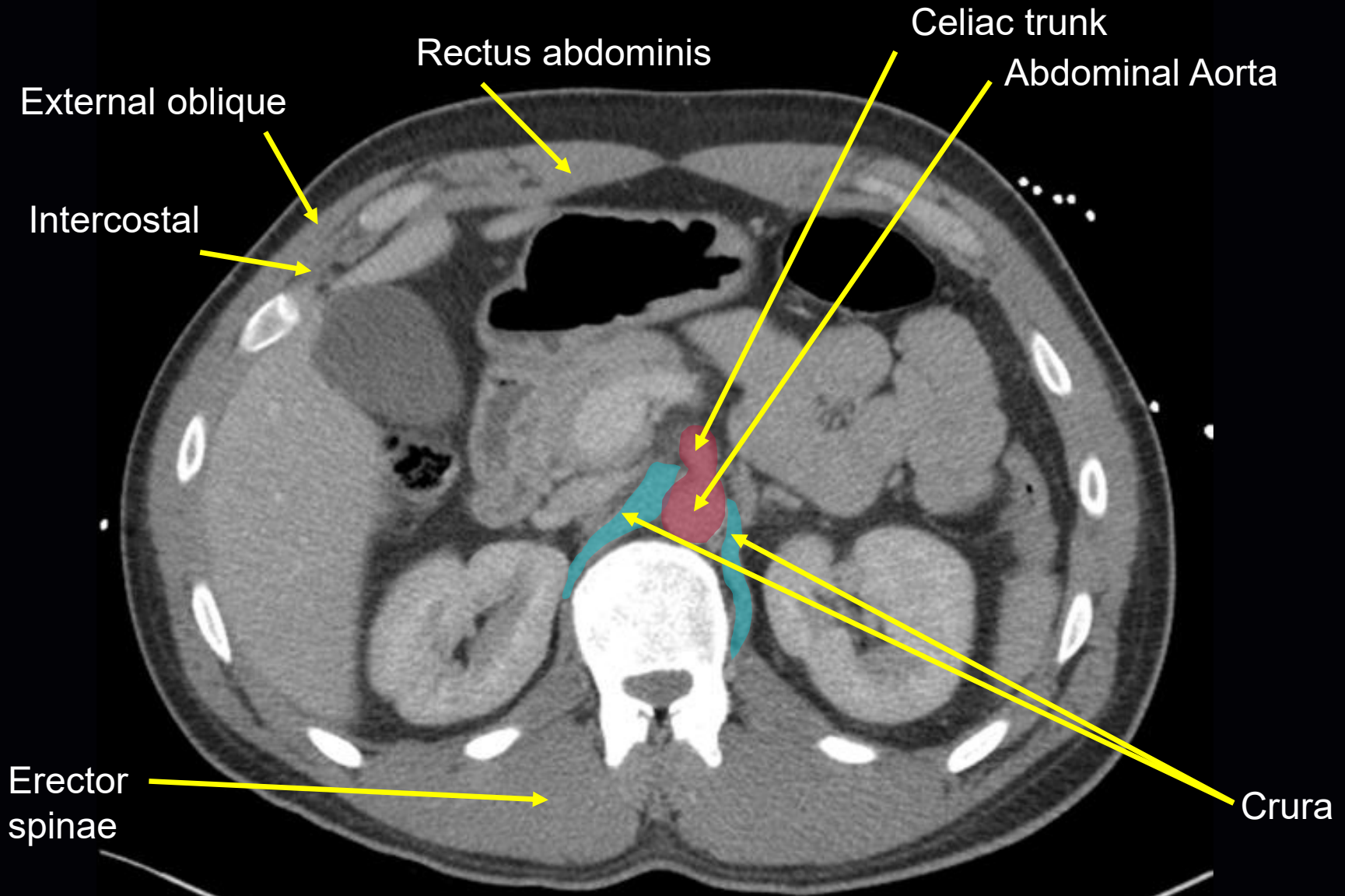
How to approach diagnostic imaging

- **Begin with the basics:** Check patient details (name, age, date of scan) /indication/ Check all slices and views (axial, sagittal, and coronal).
- **Review scan parameters:** CT/MRI; contrast/ non-contrast.
- **Start with bones:** Check vertebral column, sacrum, pelvis, femur, and ribs for any fractures, displacement, dislocation, lytic or sclerotic lesions.
- **Assess the muscles:** Look for hernias, wall thickening, or any masses.
- **Evaluate the organs:** (ER Module - BPM 501)
- **Evaluate the vascular structures:** Look at the aorta for aneurysms or dissection. Examine other large vessels for any abnormalities.
- **Evaluate the lymph nodes.**
- **Examine the soft tissues and fat planes.**

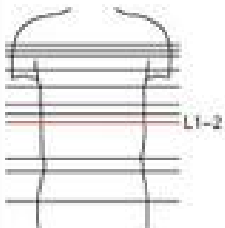
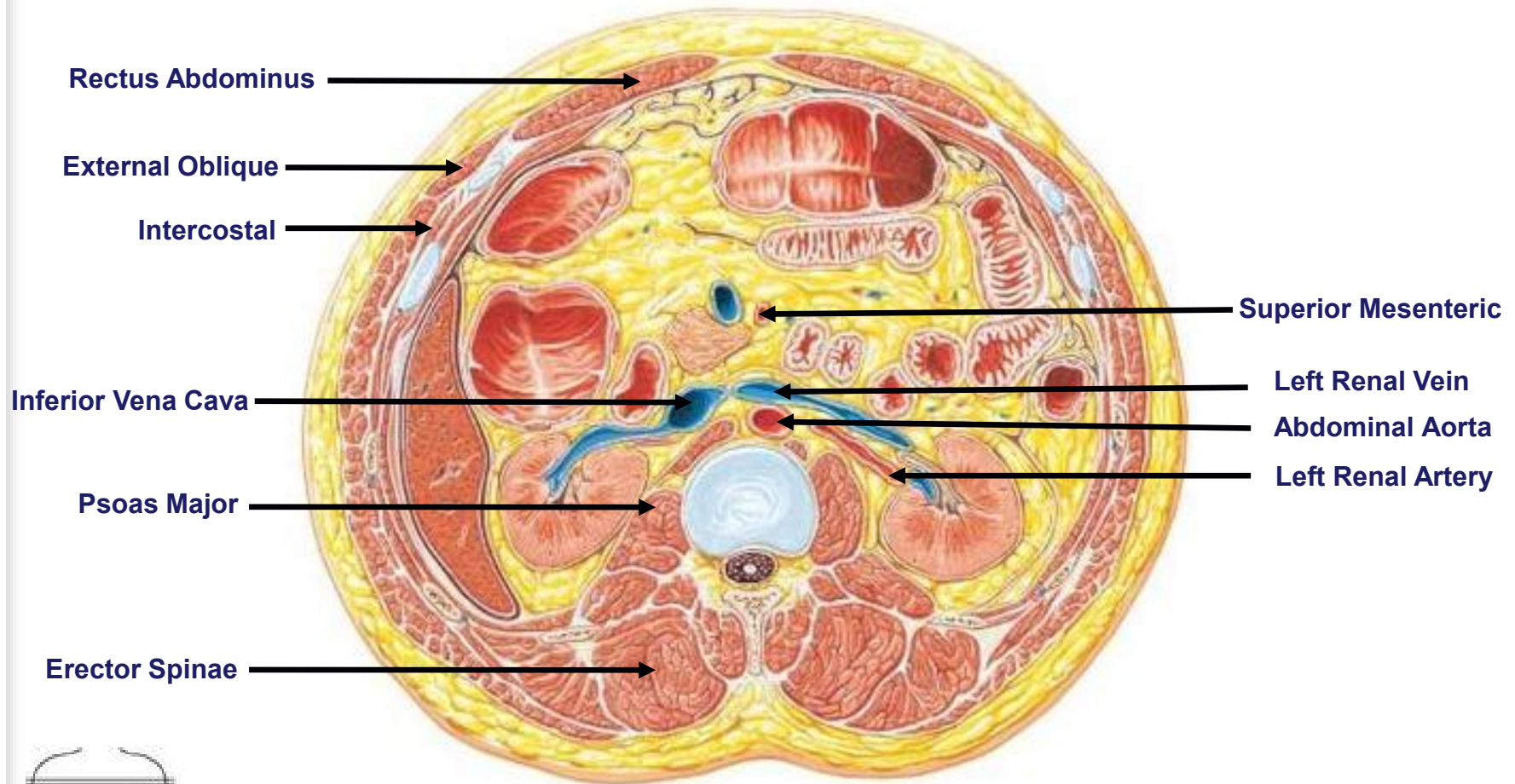
Schematic Cross Section of Abdomen at T12-L1



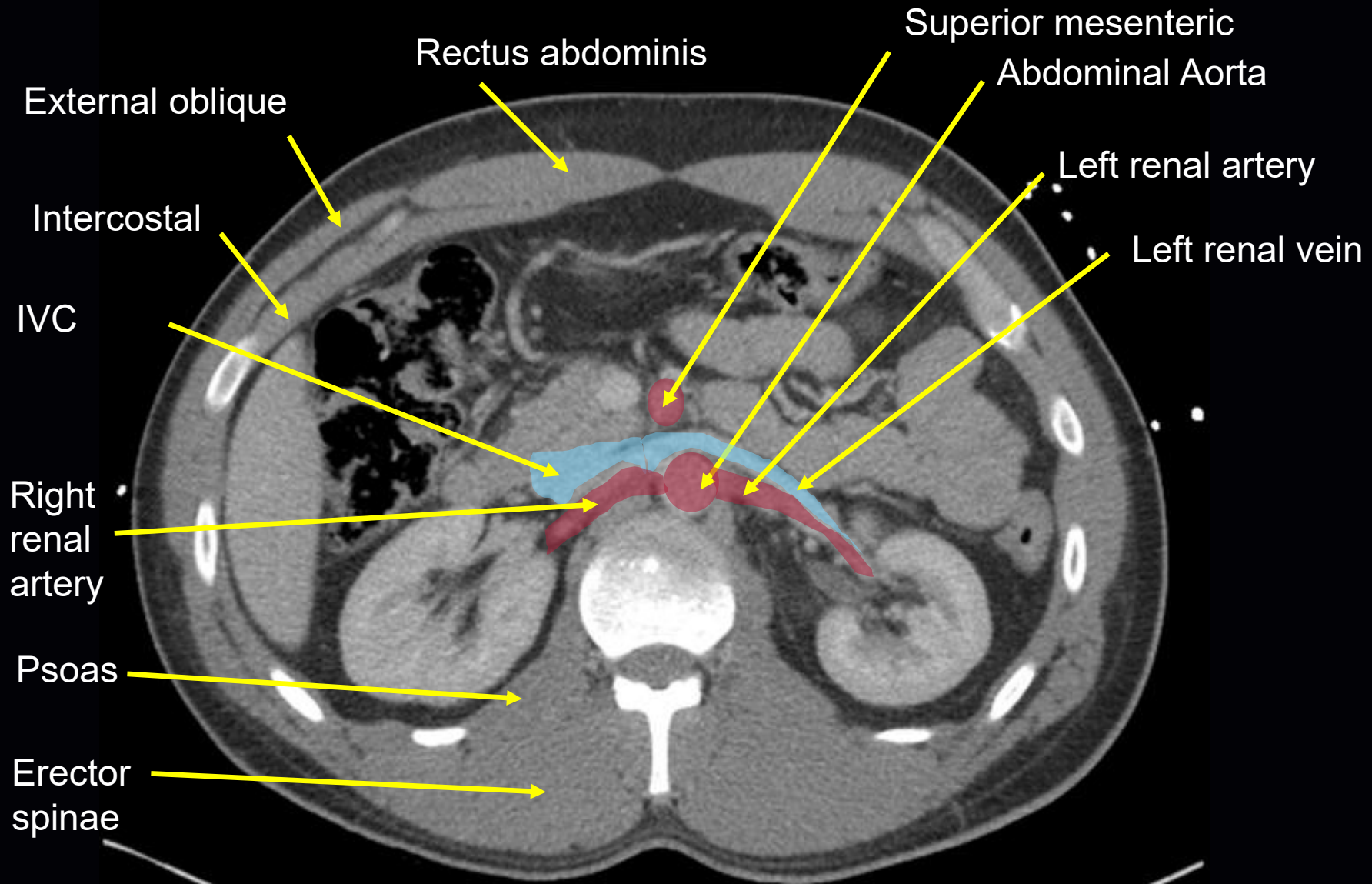
Axial non-contrast CT scan at T12-L2



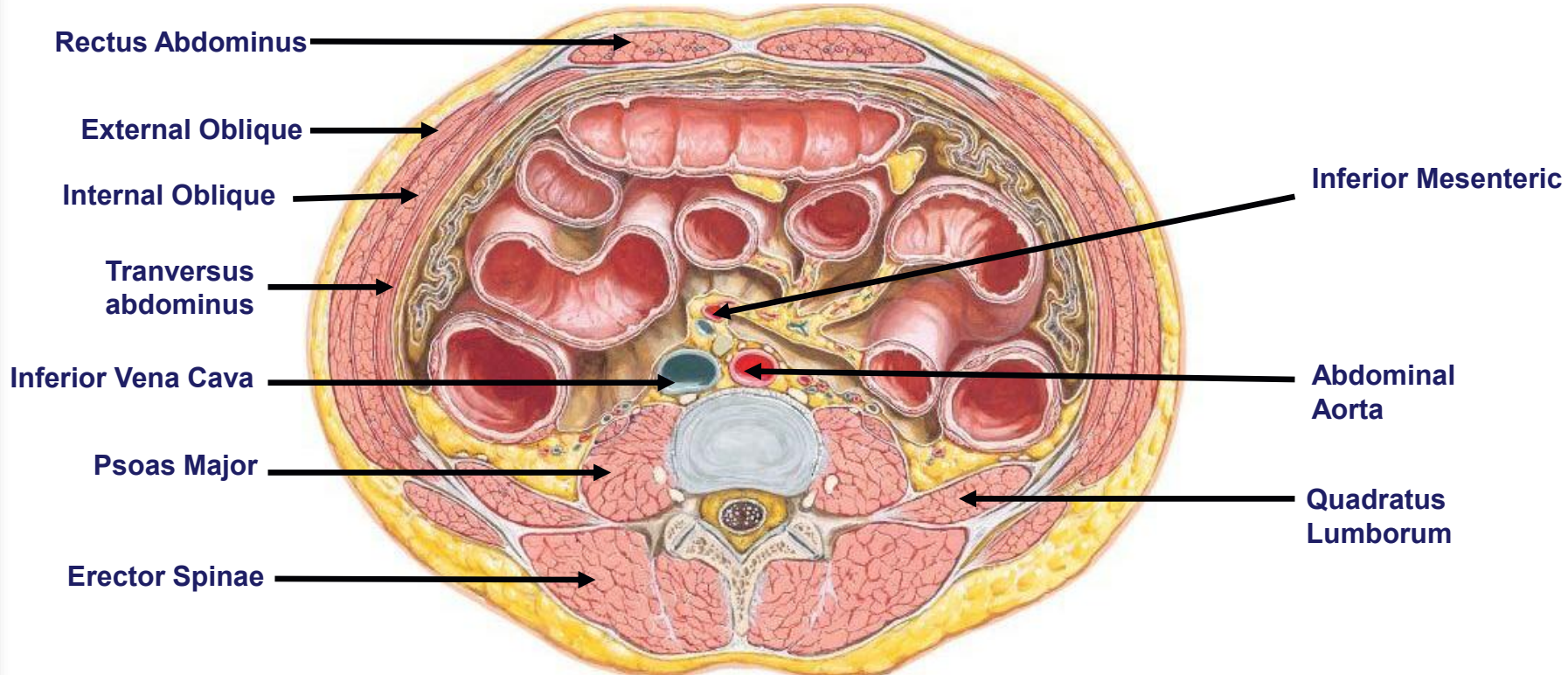
Schematic Cross Section of Abdomen at L1-L2



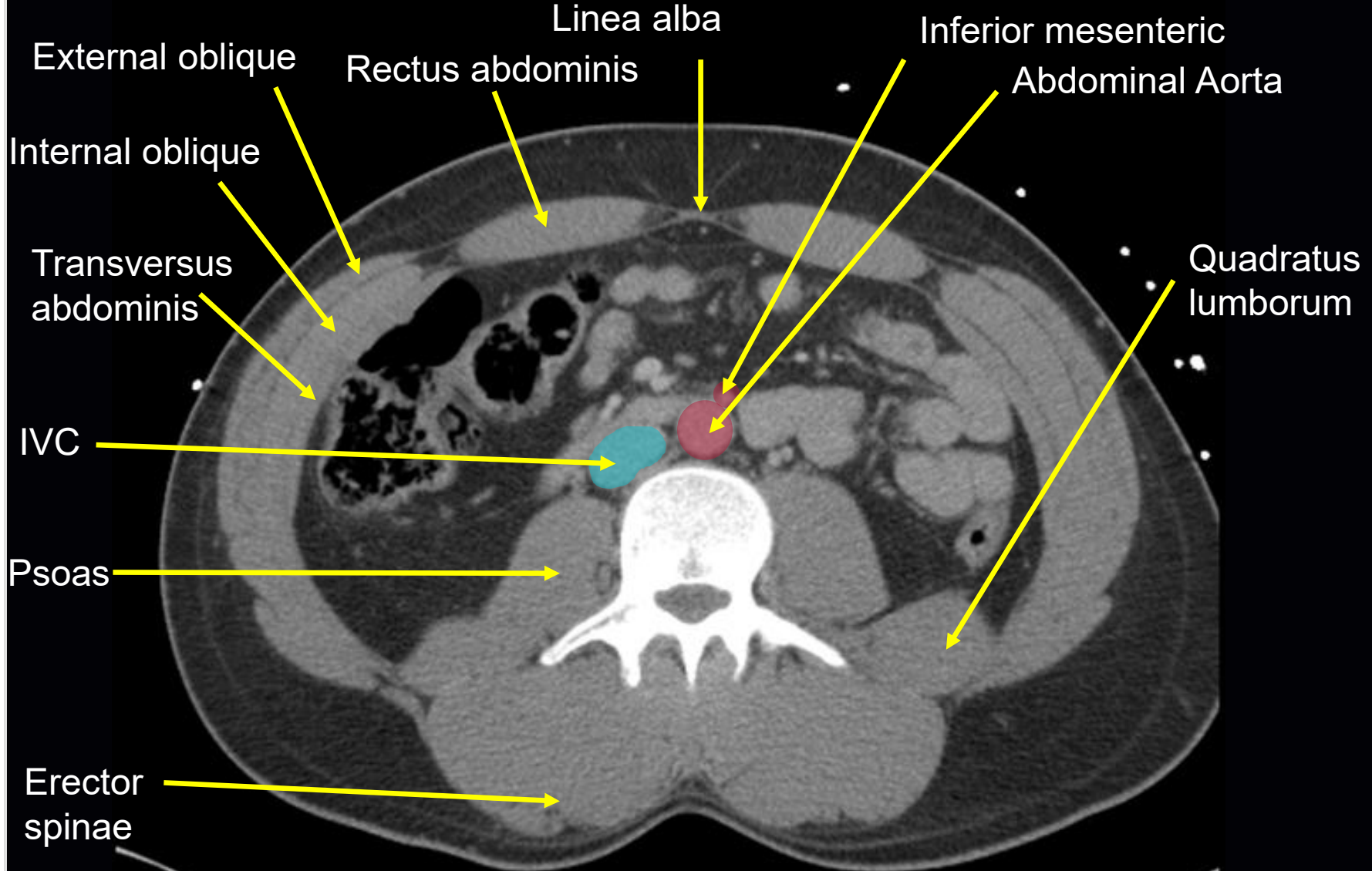
Axial non-contrast CT scan at L1-L2



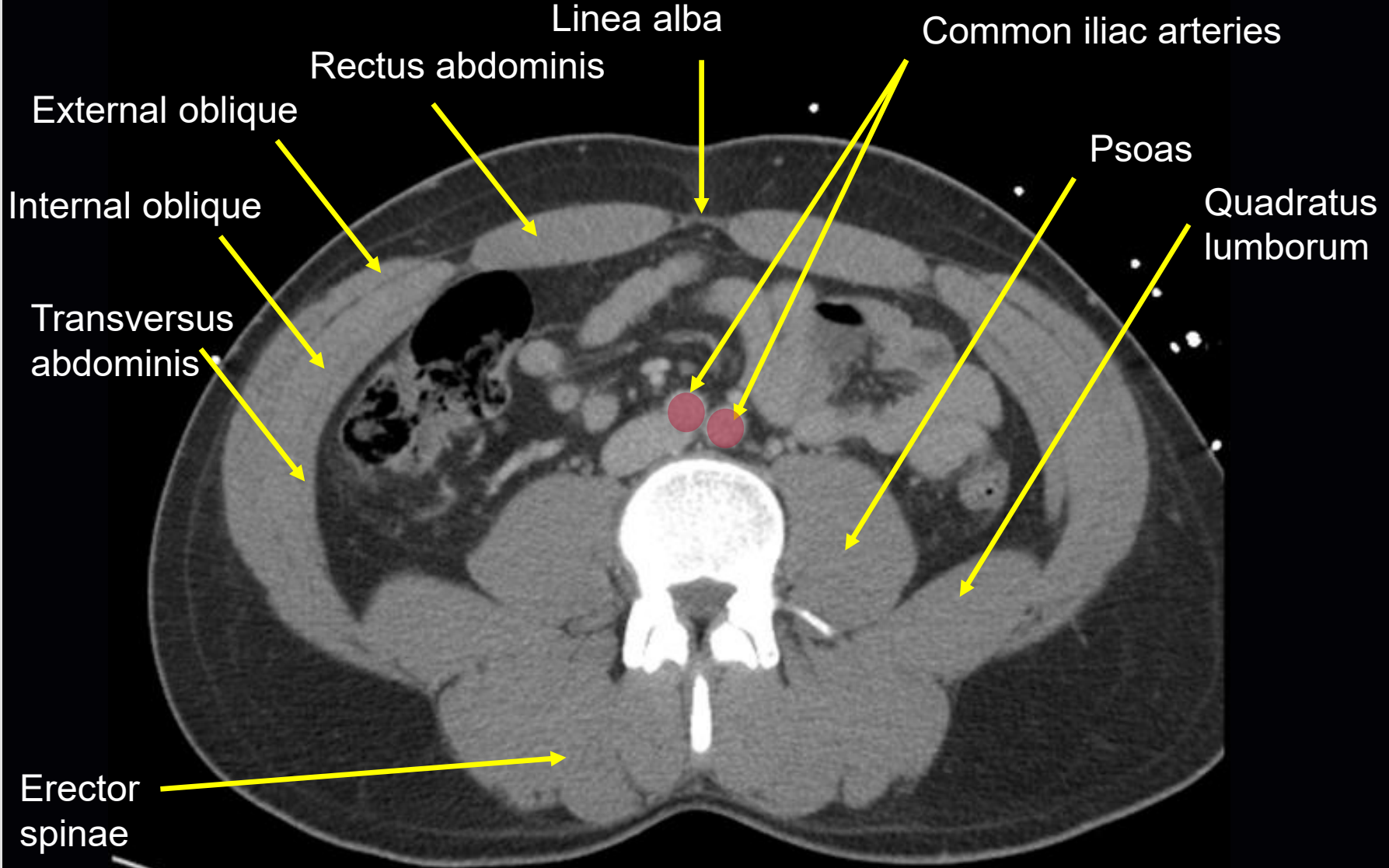
Schematic Cross Section of Abdomen at L3,4



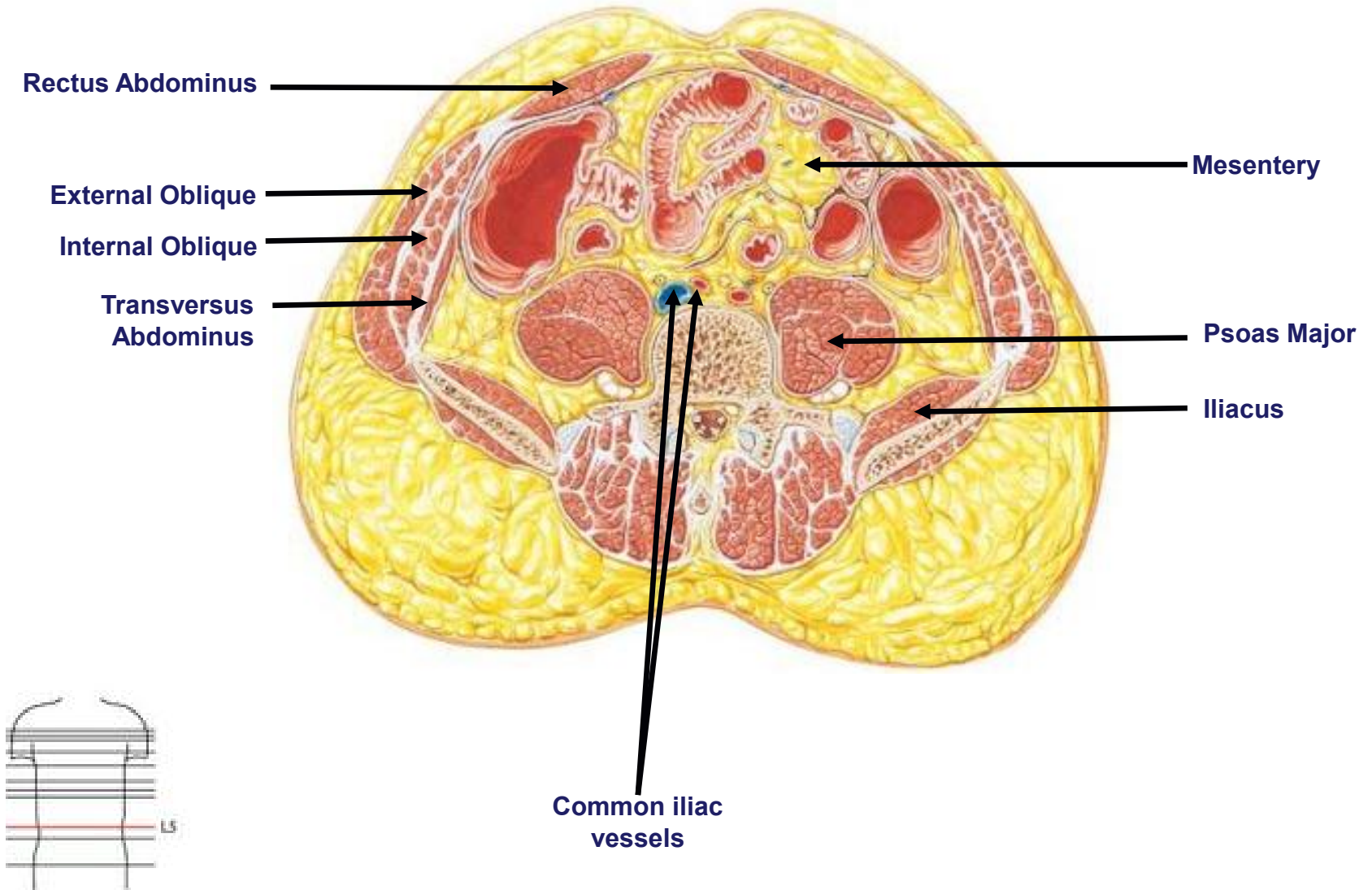
Axial non-contrast CT scan – L3



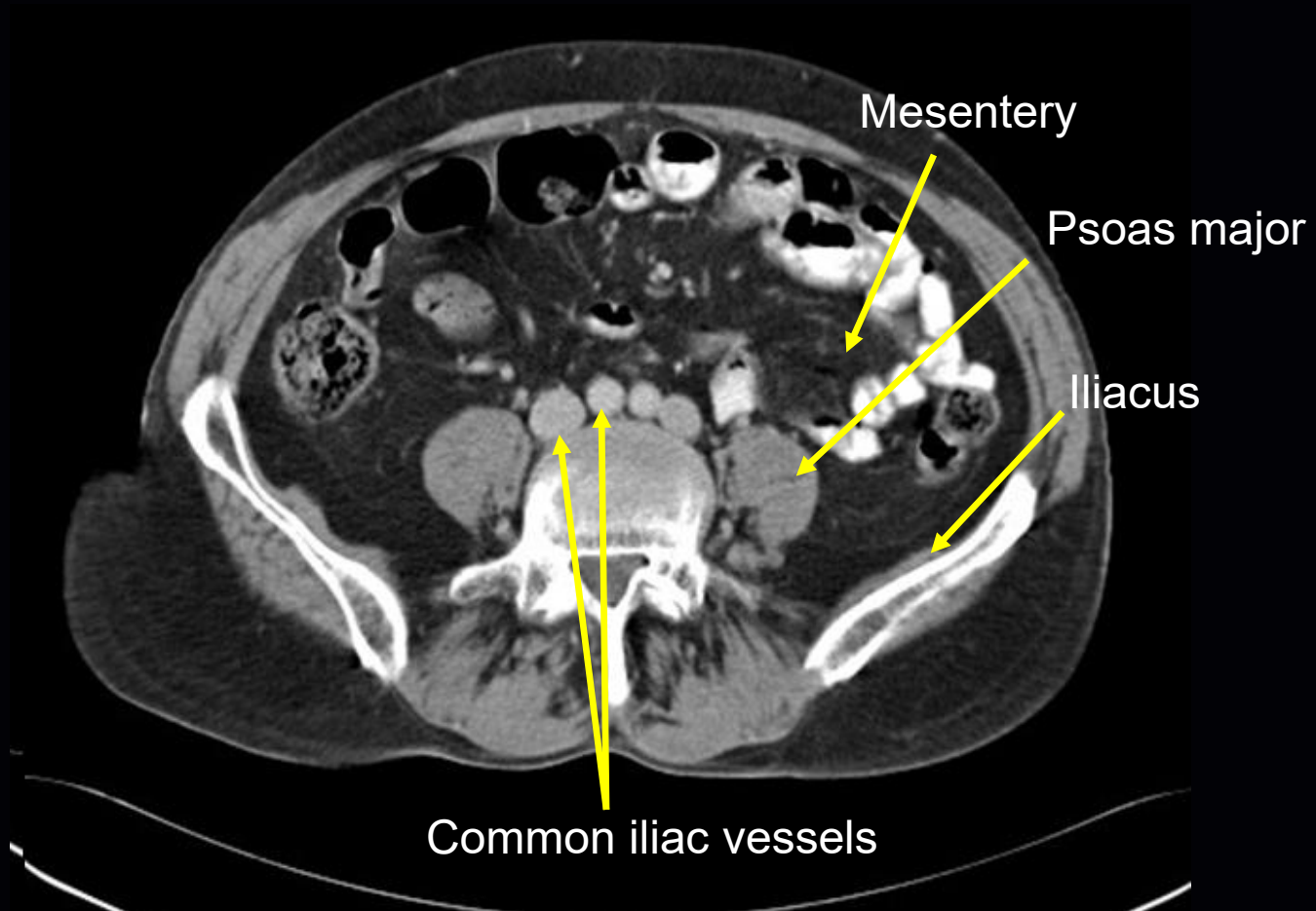
Axial non-contrast CT scan – L4



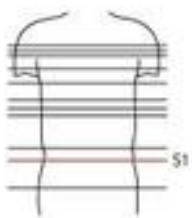
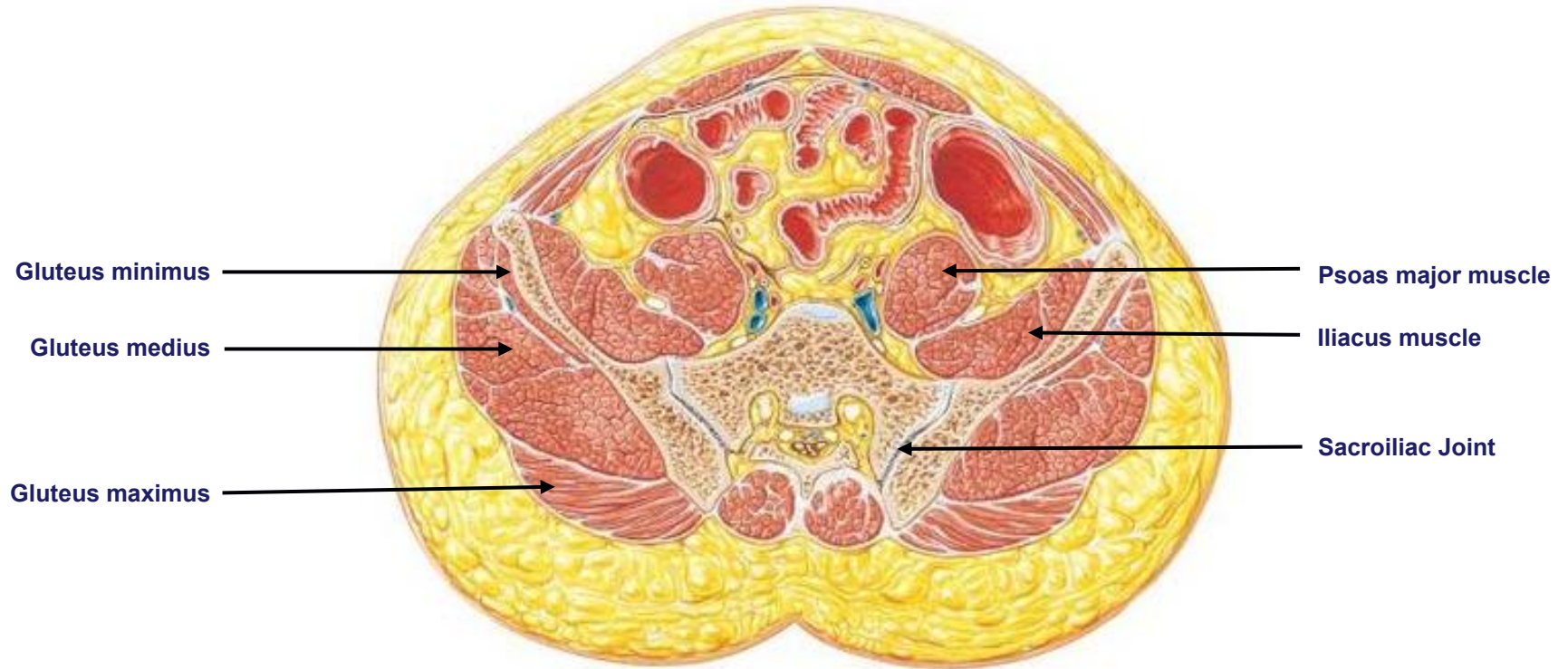
Schematic Cross Section of Abdomen at L5



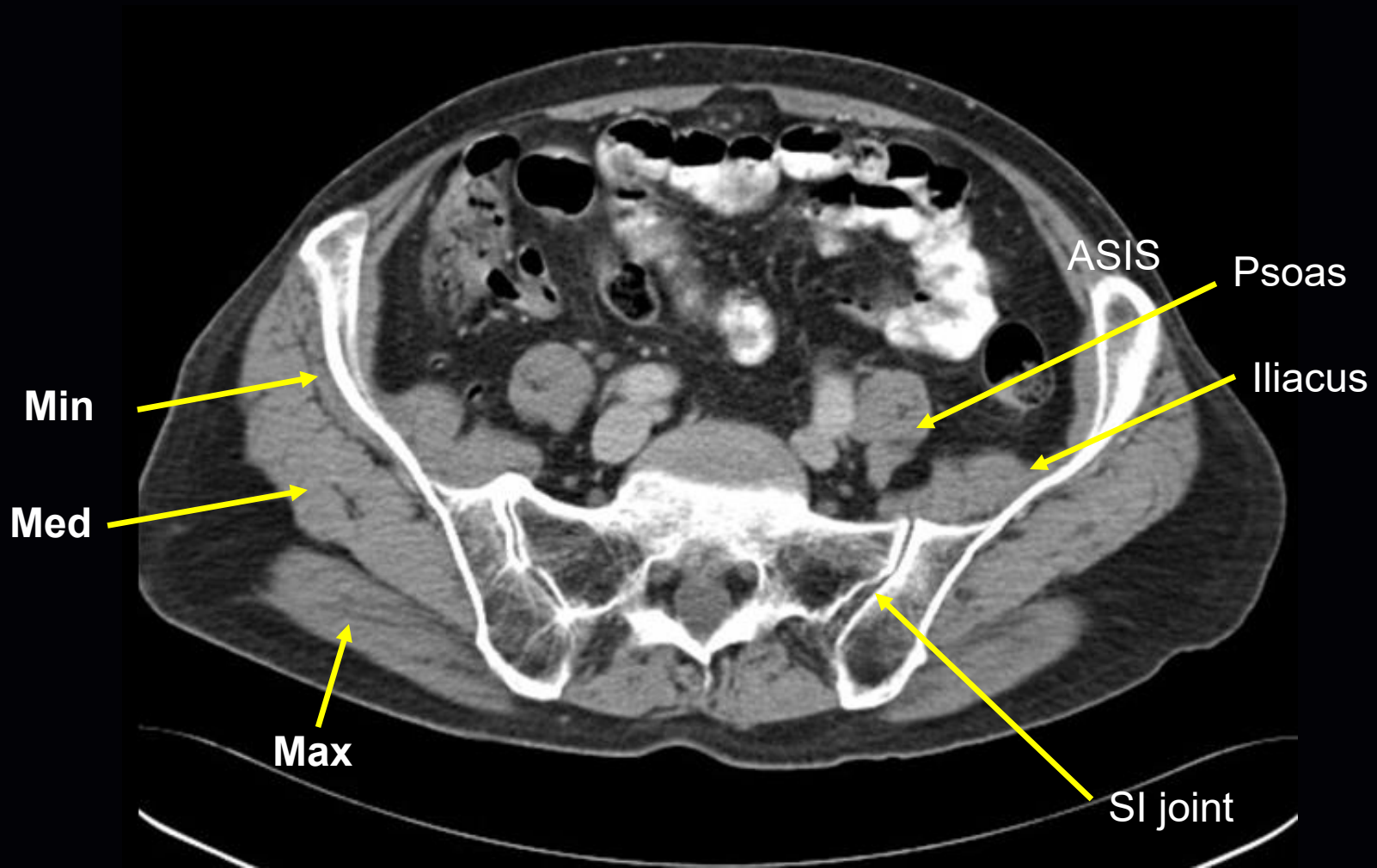
Axial non-contrast CT scan at L5



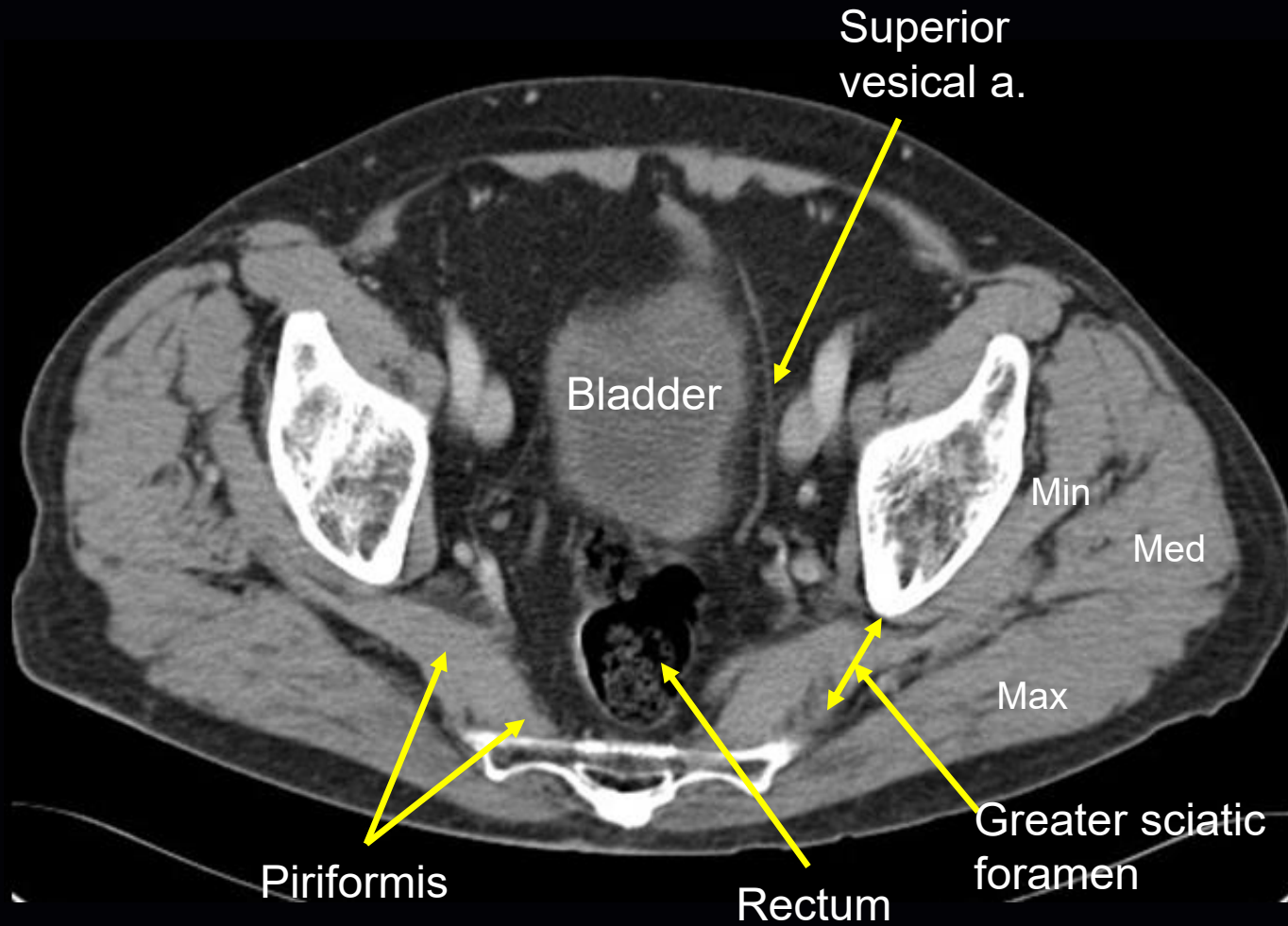
Schematic Cross Section of Abdomen at S1



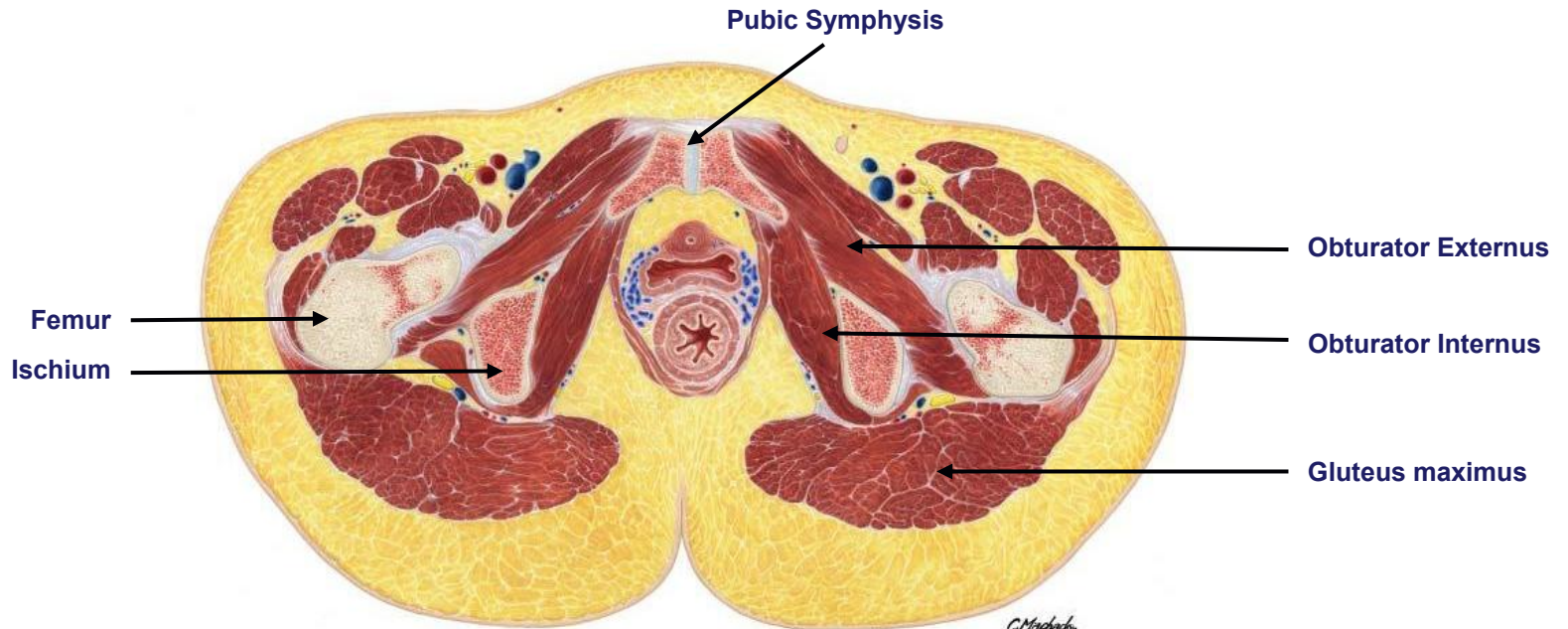
Axial non-contrast CT scan at S1



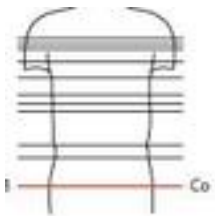
Axial non-contrast CT scan at Sacrum



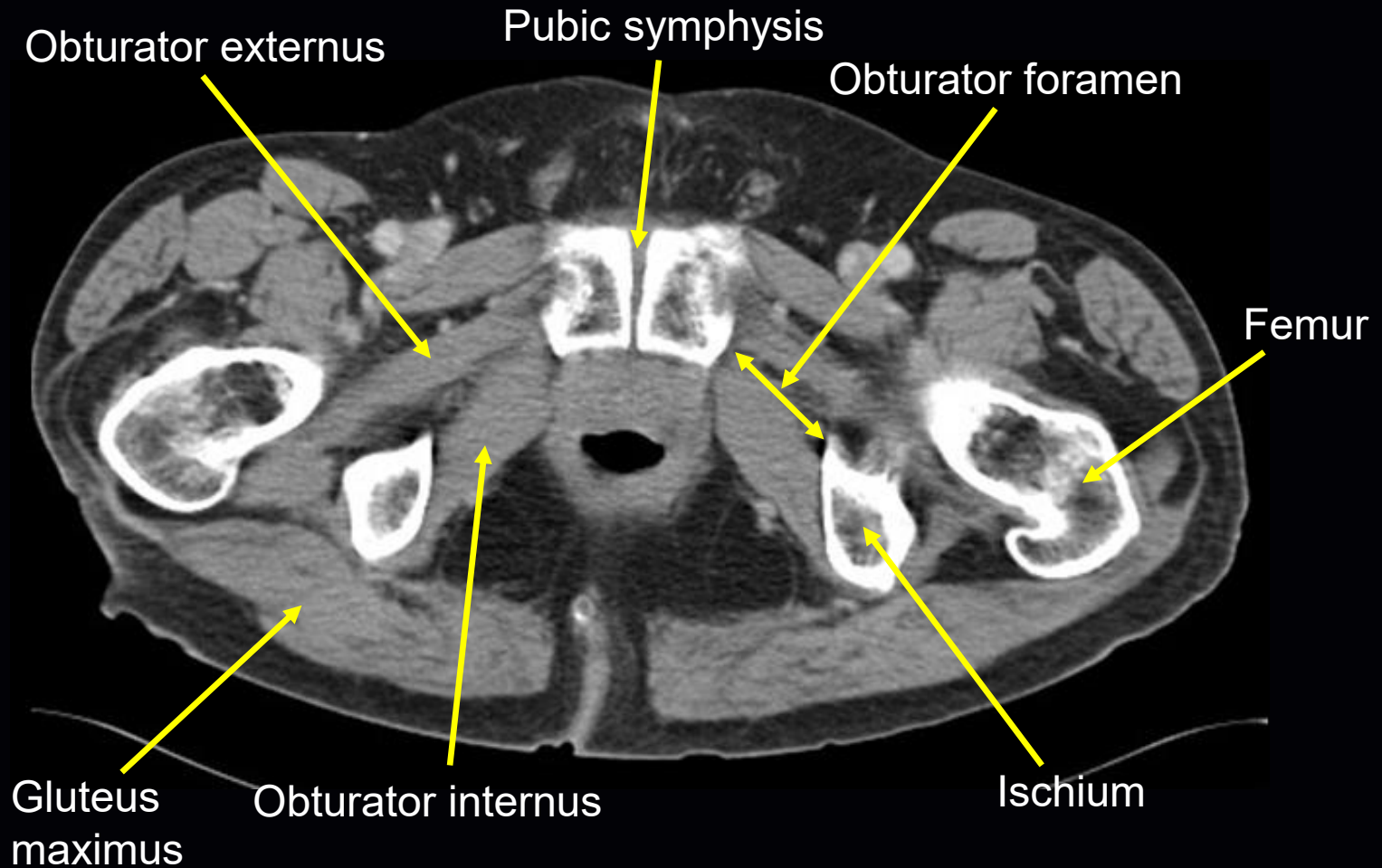
Schematic Cross Section of Abdomen at Coccyx



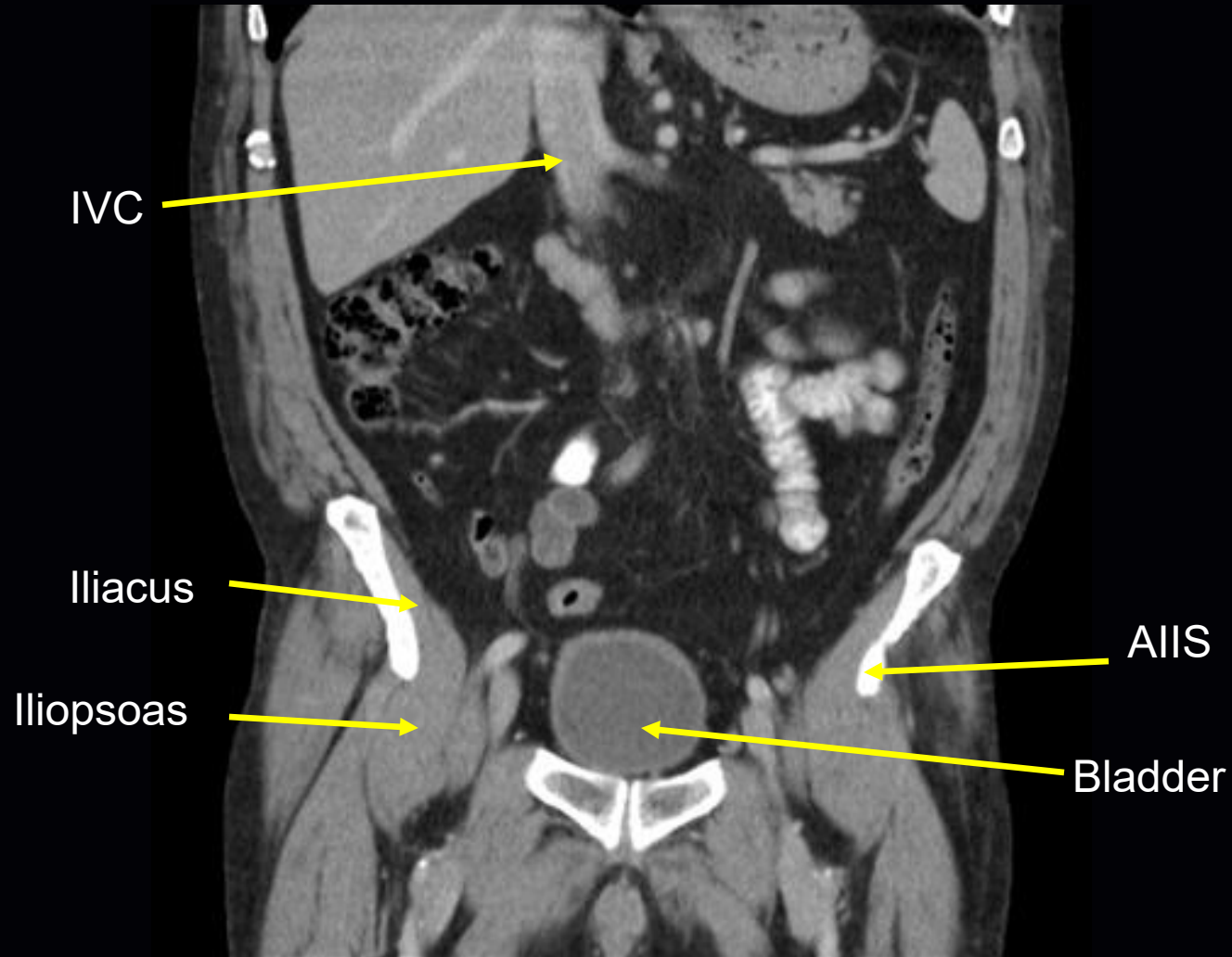
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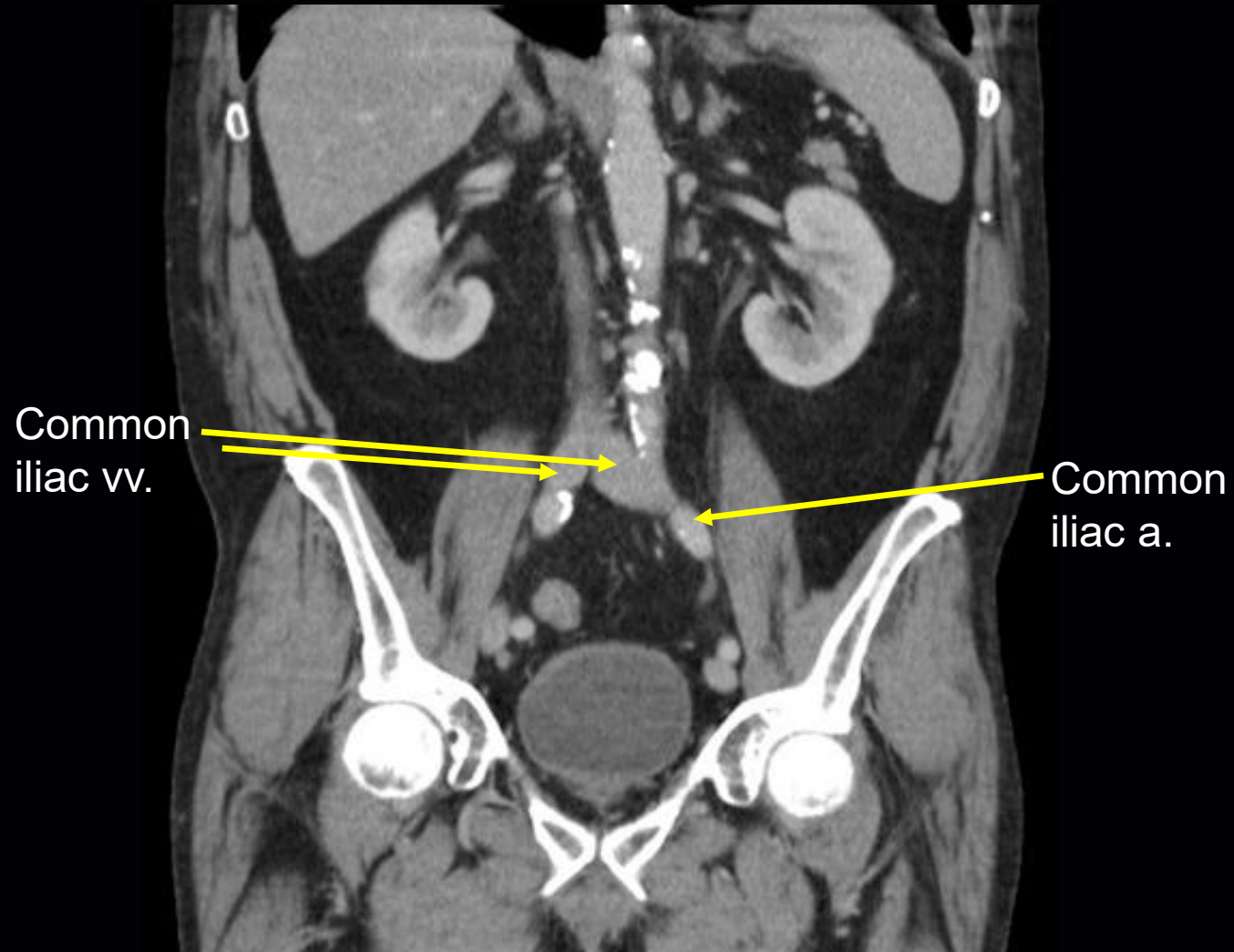
Axial CT scan at Coccyx



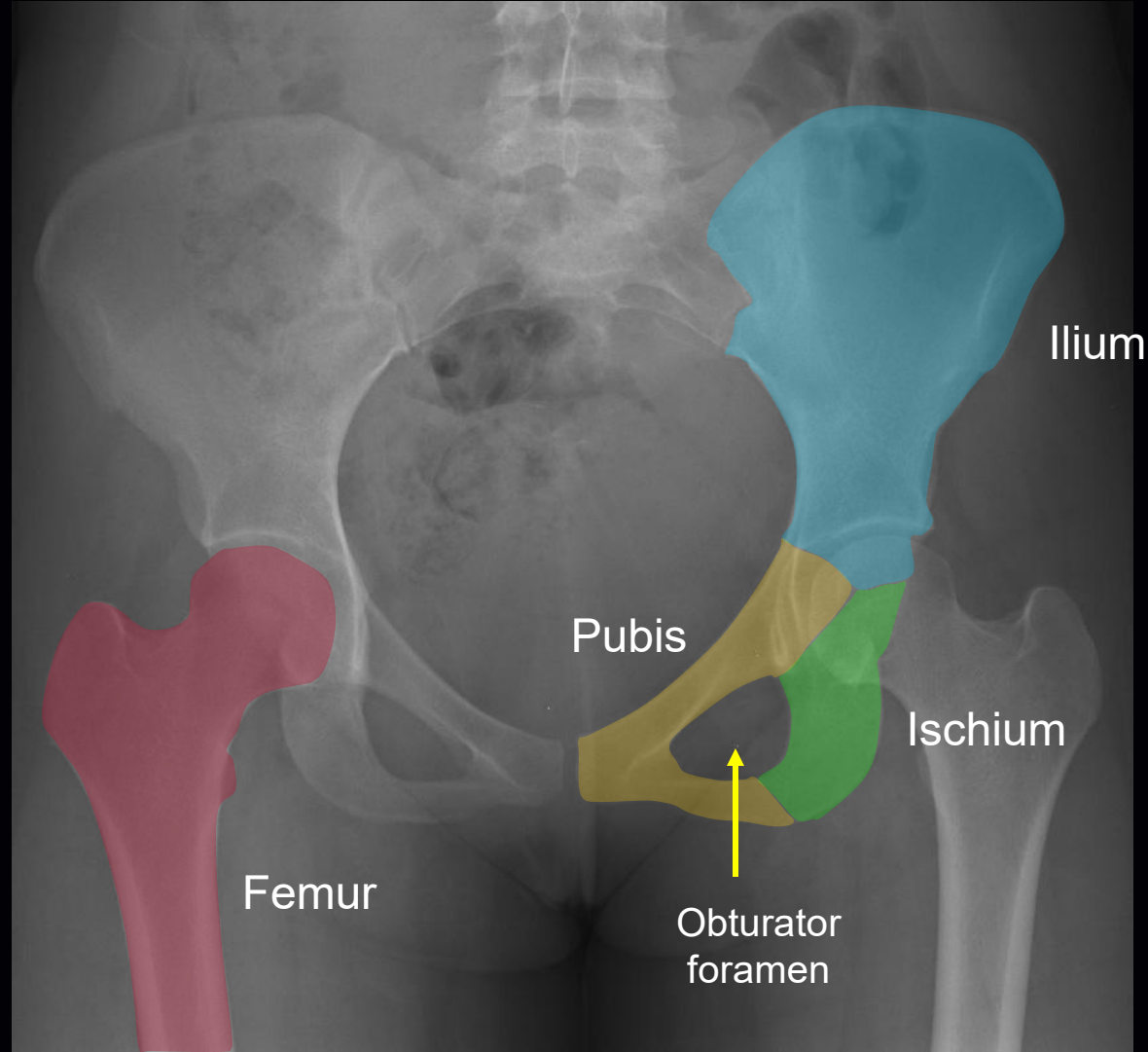
Coronal CT scan



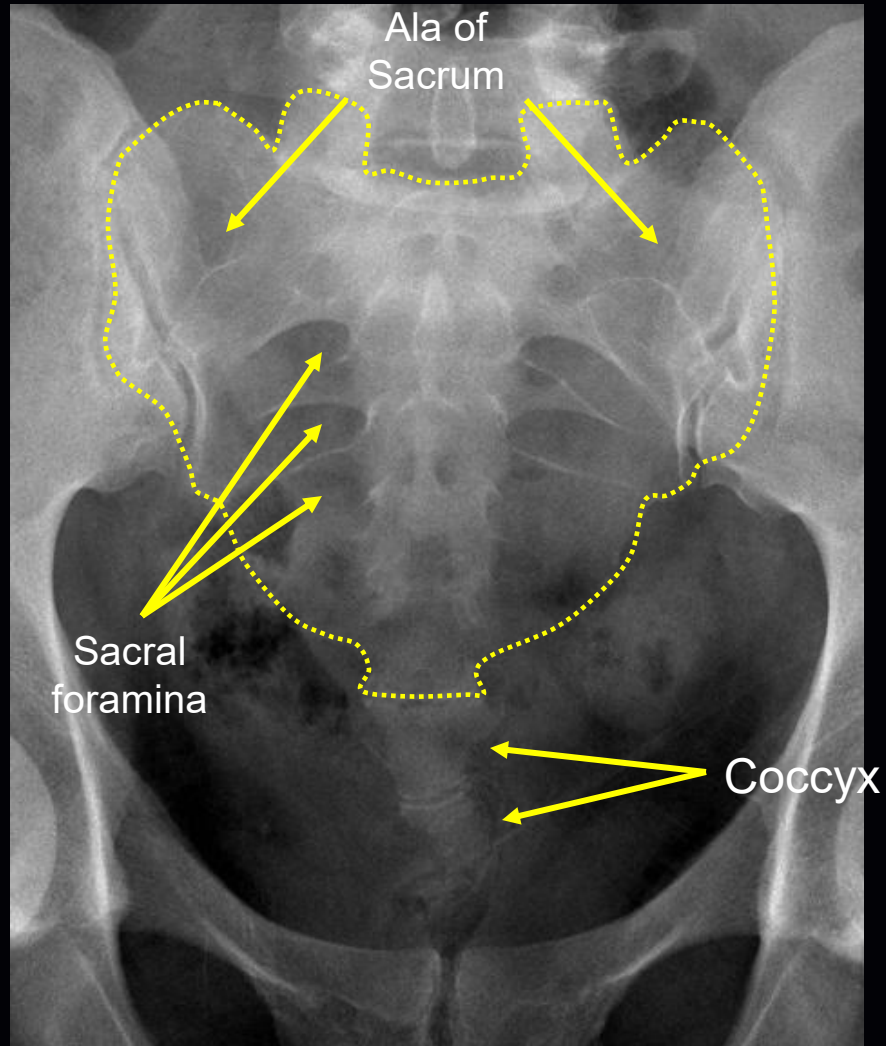
Coronal CT scan



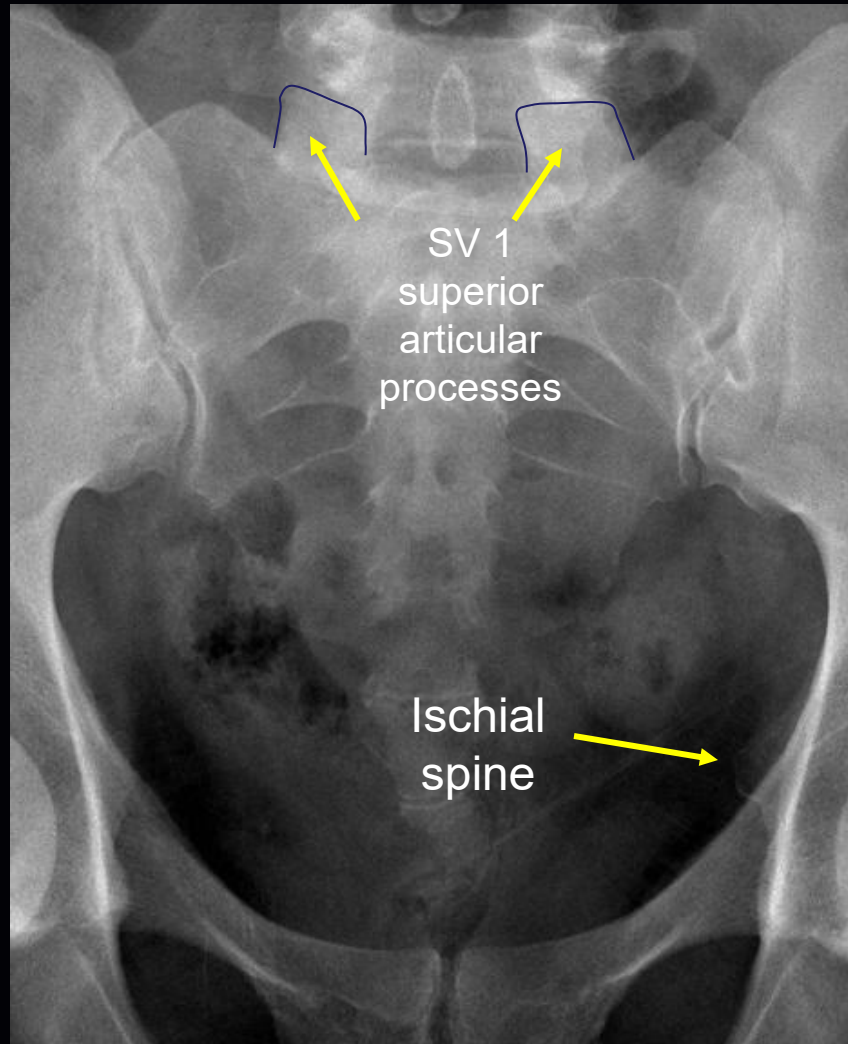
AP Radiograph of Pelvis



AP Radiograph of Pelvis



AP Radiograph of Pelvis

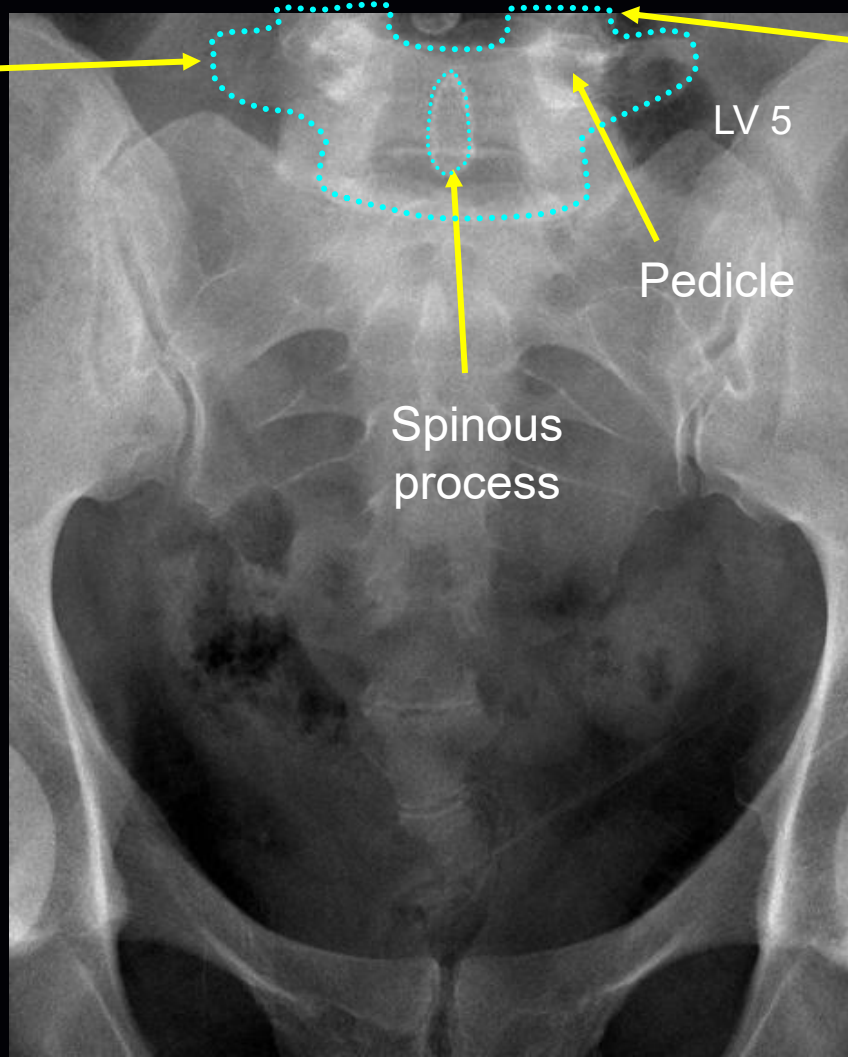


Sacrum

AP Radiograph of Pelvis

Transverse
process

Superior articular
process

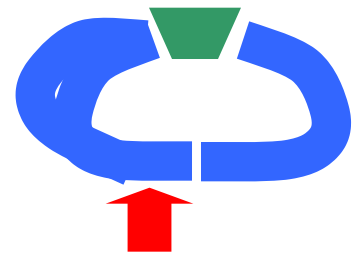
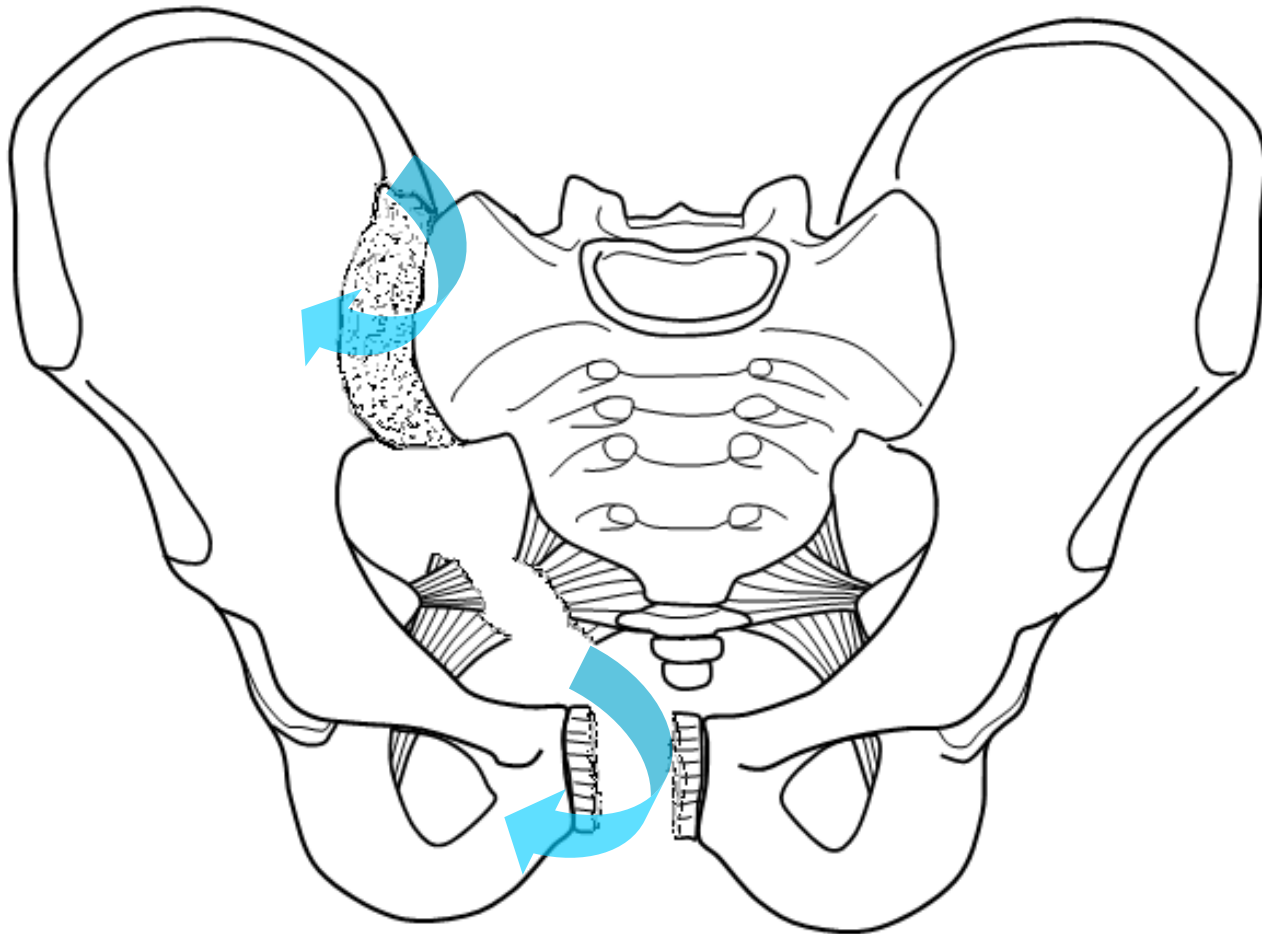


Pelvic Fractures

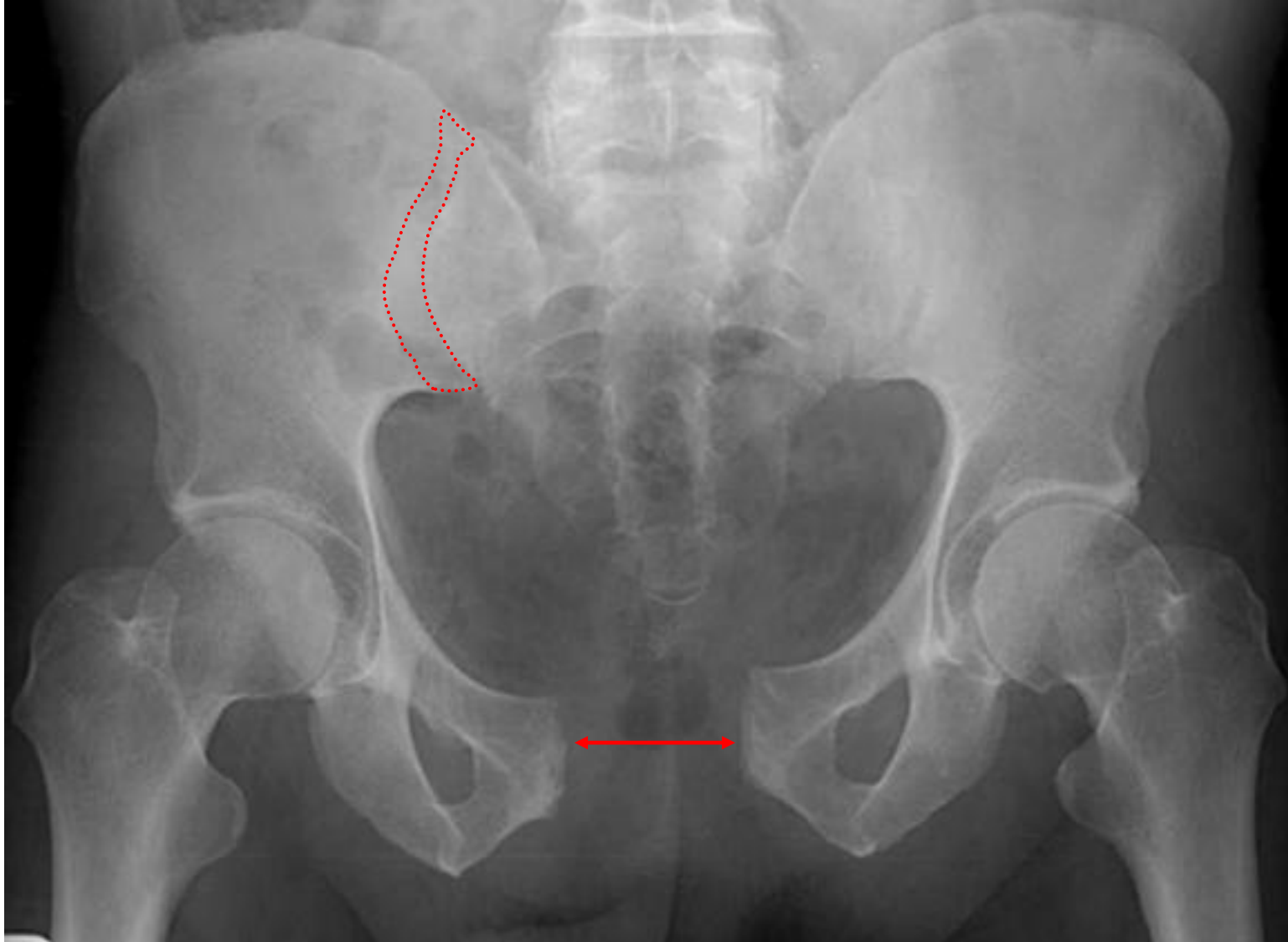
- Pelvic fractures commonly result from “high-energy” trauma (motor vehicle collisions, falls).
- These fractures have significant associated morbidity and mortality arising from:
 - Complications of the fractures: Pelvic hemorrhage.
 - Commonly associated injuries (such as head injuries, etc., that can also occur in high-energy trauma).
- Single fractures / ligamentous disruptions of the bony pelvis are uncommon (Except in small children, where there is a high elasticity of bones/ligaments).
- Mechanisms:
 - Anterior-Posterior Compression
 - Lateral Compression
 - Vertical shear injuries

Anterior-Posterior Compression

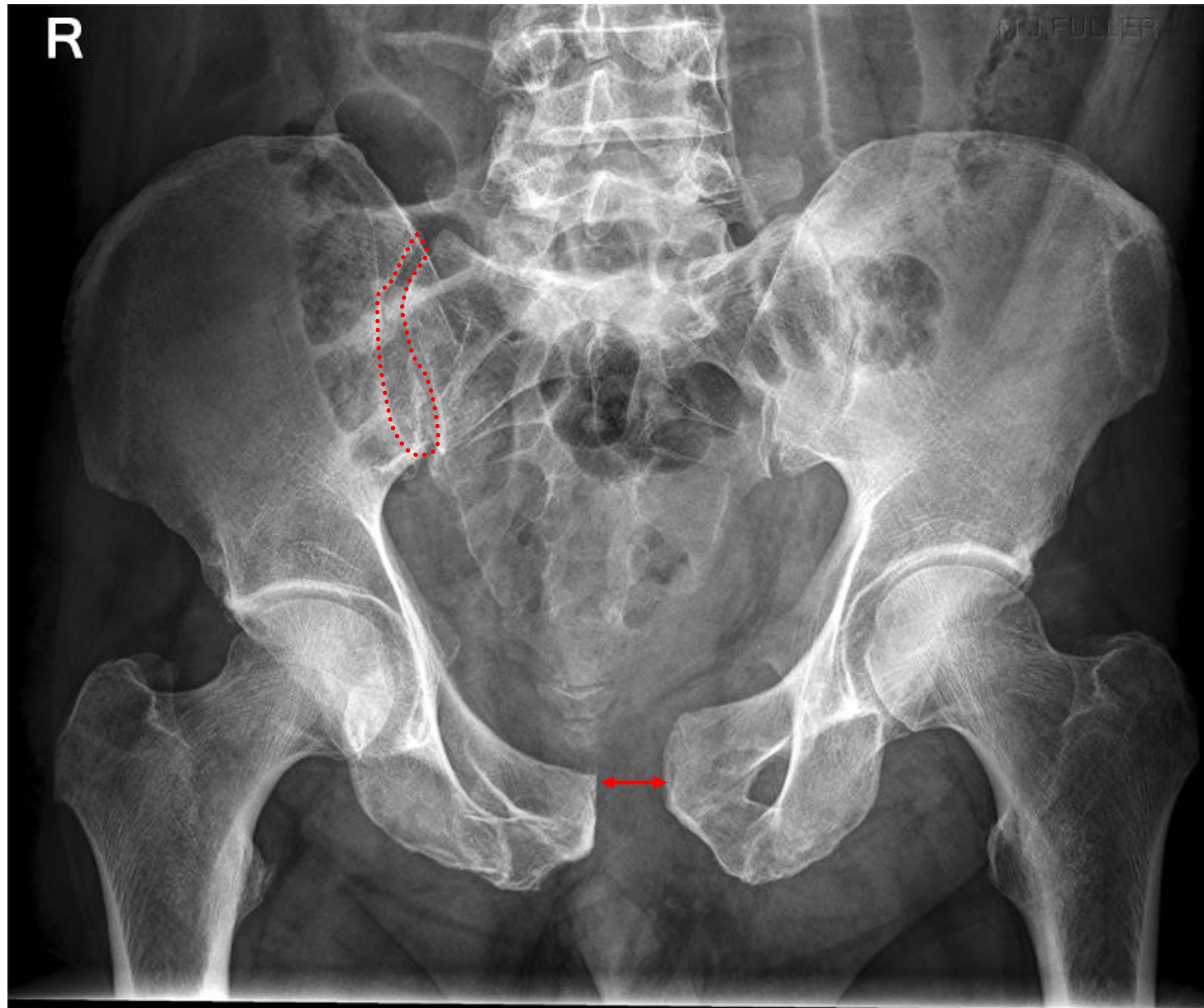
- AP Compression (“Open Book”) – diastasis at pubic symphysis and R sacroiliac (SI) joint.



Anterior-Posterior Compression



Anterior-Posterior Compression

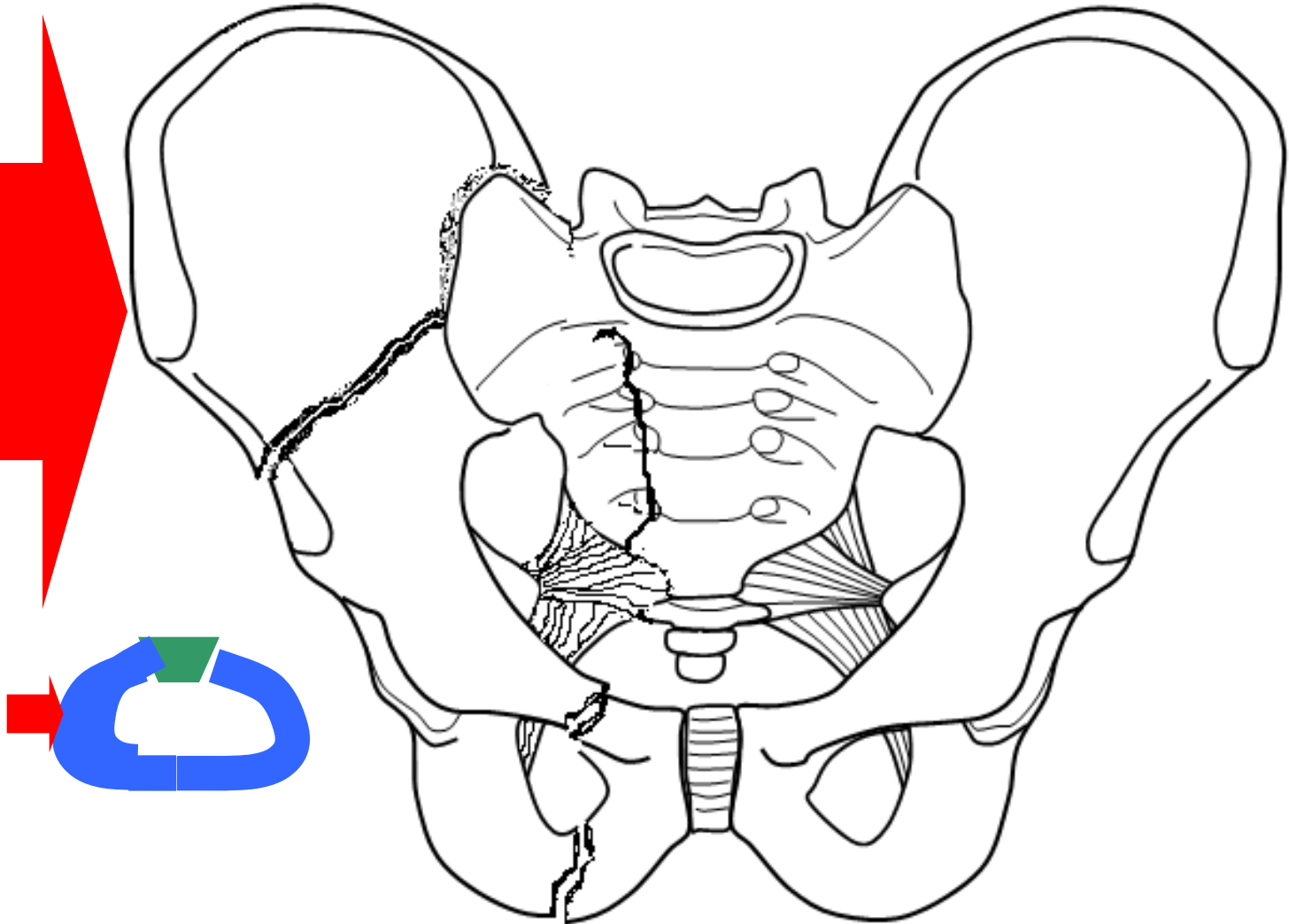


Anterior-Posterior Compression

- Most commonly result from **head-on motor vehicle collisions** (can also occur with motorcycle collisions or when pedestrians are struck by a vehicle).
- Initially, the **pubic symphysis is disrupted** (either through the fibrocartilage fibers or through a rim of pubic bone on one side or the other).
- The **anterior sacroiliac ligaments** are affected **first**, but with sufficient force the posterior ligaments can also be disrupted.
- When sufficient force is present, the **sacrospinous** and **sacrospinous** ligaments are disrupted.
- The hemipelvis (one or both) is externally rotated (**opened like a book**).

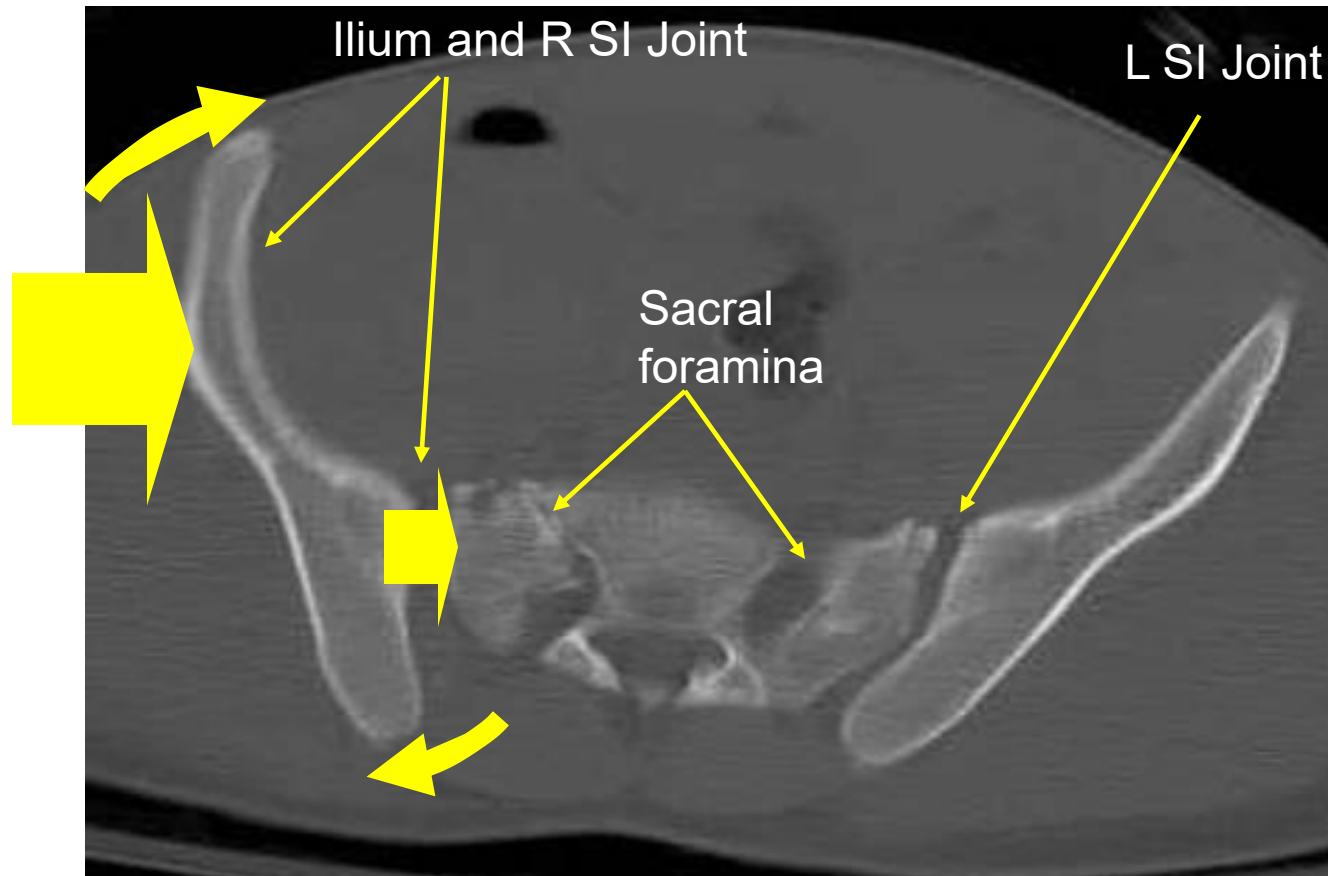
Lateral Compression

- Lateral Compression – sacral “buckle” (impacted) fracture, ” fractures of pubic rami, iliac wing



Lateral Compression

- CT – right lateral compression injury – internal rotation of R hemipelvis and R sacral buckle fracture

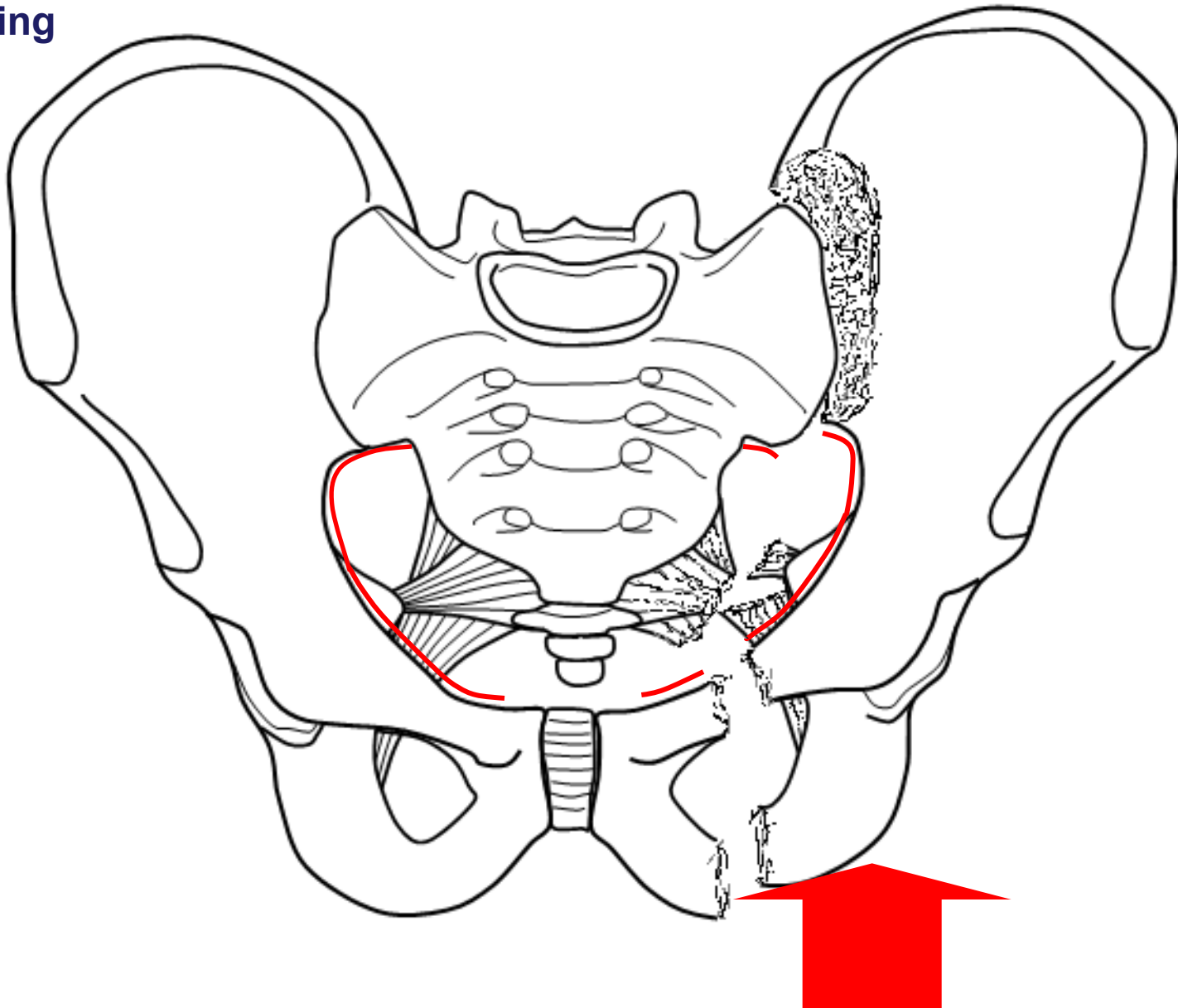


Lateral Compression

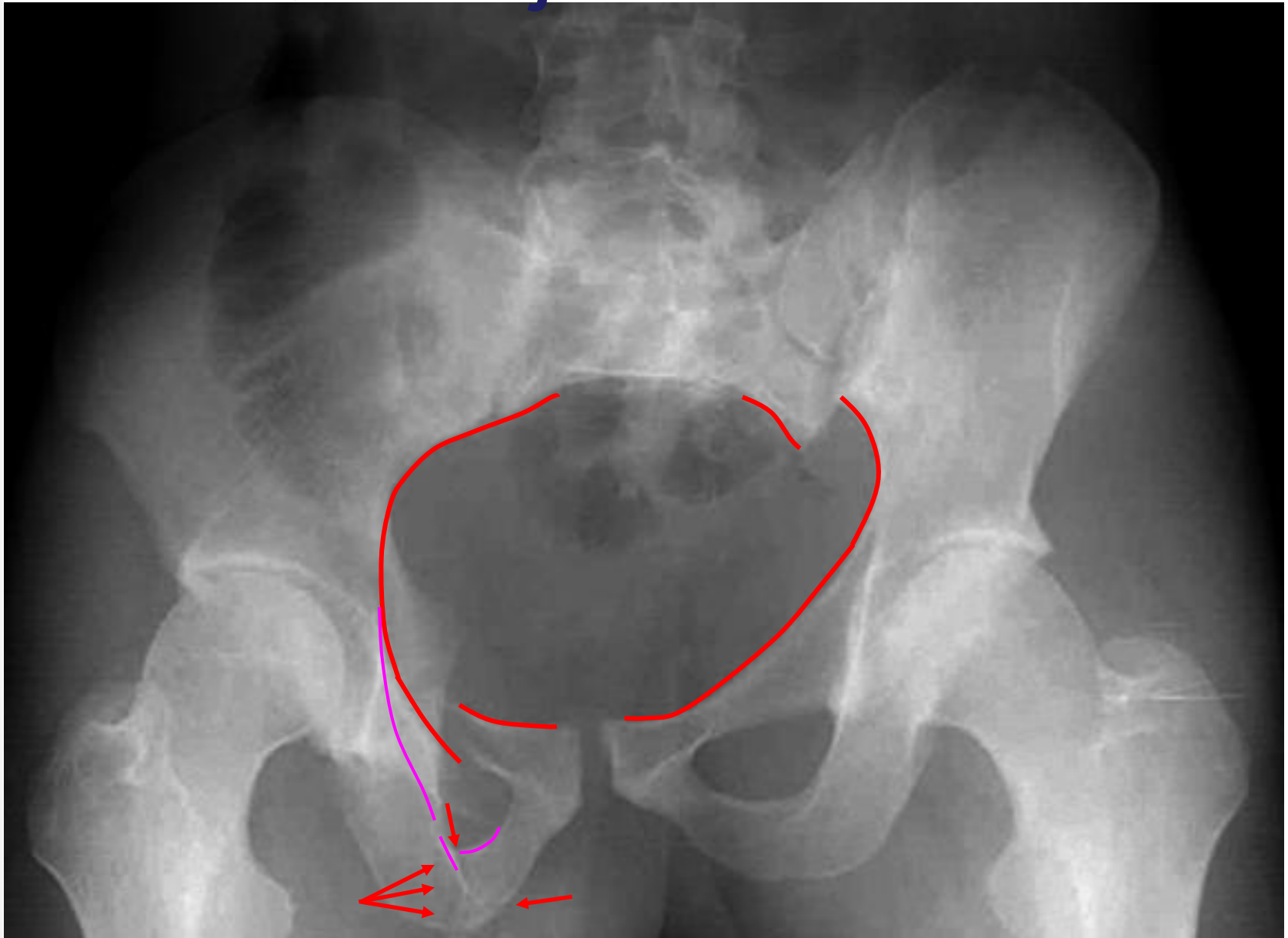
- Lateral Compression - result from **side impact** such as pedestrian vs. vehicle or side-impact motor vehicle collisions.
- The **sacrum “buckles”** (impacted fracture - typically on the side where the force is applied) and the **iliac wing** and **pubic rami fracture** (one side or both) as **the hemipelvis is forcefully rotated internally**.

Vertical shear injuries

- sacral “buckle” (impacted) fracture, ” fractures of pubic rami, iliac wing



Vertical Shear Injuries



Vertical shear injuries

- Vertical shear injuries - **fall from a height on to one (or both) “outstretched” lower limbs**, can also occur in motor vehicle collisions (knee – dashboard). The force is directed from inferior to superior (caudo-cranial).
- The force drives one (sometimes both) **hemipelvis “upward”, posteriorly disrupting the SI joints**, disrupting the **sacrospinous and sacrotuberous ligaments**, and anteriorly disrupting the pubic symphysis (or pubic rami, ipsilateral, contralateral or both).

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